

oscNext - Simulations and sample (v00.07)

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Abstract

This document describes the ‘oscNext’ suite of atmospheric neutrino oscillation measurements using DeepCore data from 2011-20. This is a large group effort involving many institutions and authors from the oscillation working group.

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1 Executive summary

1.1 General goal of the analysis

‘oscNext’ refers to a suite of atmospheric neutrino oscillation analyses (both conventional and BSM) using all available DeepCore data (2011-2019). This effort includes new:

- Simulations (neutrinos, muons, noise)
- Event selection (muon and noise rejection)
- Reconstruction methods
- Particle ID (PID) methods
- Systematic uncertainty modelling

The primary goal of this technical note is to provide all the necessary information to understand all these elements. The other goal is to provide a detailed reference for future analysers for the next generation of analyses and to facilitate reproducibility. The heavier technical details can be found in the appendices.

1.2 Simulation

Three classes of events are present in DeepCore data and must be simulated:

- Neutrinos (GENIE, 1 GeV - 10 TeV)
- Atmospheric muons (MuonGun)
- Randomly coincident noise (vuvuzela)

All simulations are produced by the oscillations working group, and are designed to match pass 2 detector data (charge calibration, filter versions). For neutrinos, the simulated livetime is $\sim 10\times$ the detector livetime. Muon and noise events cannot be produced at these same quantities due to the vast rates at low levels, with $\sim 1\times$ the detector livetime being produced for muons and ~ 1 month for noise. The neutrino and muon simulations specifically target events in the DeepCore region (although the full detector response for these events is simulated), and a pre-scale technique is used to enhance the muon simulation efficiency in the parameter space relevant for the final level sample. No data-driven templates are used, simulation is used in all cases.

1.3 Event selection

The atmospheric muon and randomly coincident noise event rates dominate over neutrinos at filter level, and a multi-stage event selection is used to suppress the muon and noise background rate by a factor of $\mathcal{O}(10^5)$, giving a neutrino dominated sample at final level. The selection targets neutrinos starting in DeepCore (see section B for fiducial volume definition), and includes both cascade and starting track topologies. The sample includes neutrinos from all directions (all-sky), but note that some analyses choose to cut away the down-going region to remove muon background further (as is the case for the ‘verification’ sub-sample).

The methods used build upon the previous generation of oscillation analyses, and exploit state-of-the-art machine learning methods to achieve lower background and higher signal rates than previous analyses. Overall, the increase in detector livetime and improved selection result in $\sim 5\times$ more neutrinos than the previous generation of analyses.

The event selection stages are:

- L2 - common collaboration filter level (we use the DeepCoreFilter)
- L3 - simple cuts to remove ‘easy’ noise, muon and unsimulated events and achieve good data-MC agreement before machine learning
- L4 - machine learning (BDT ensembles) to remove noise and muons
- L5 - removal of muons entering DeepCore via lightly-instrumented ‘corridors’ in the IceCube veto
- L6 - final level neutrino reconstruction (starting track+cascade hypothesis)
- L7 - final level muon and noise rejection cuts using final level reconstructions

The event sample splits at L6 into two distinct samples; a *verification sample* with lower statistics but high-quality up-going events and a simple reconstruction algorithm, that can readily be used to investigate systematic effects and test the analysis tools more rapidly, and a *high-statistics sample* that uses the full power of the statistics accumulated in the detector to make more sensitive measurements of neutrino oscillation. There are thus two distinct L6 and L7 stages, although they follow the same objectives outlined above.

The final level samples have $\mathcal{O}(1\text{ MHz})$ neutrino rates, a few % muon contamination and negligible noise contamination.

1.4 Reconstruction and PID

Final level neutrino reconstruction utilises a starting event cascade+track hypothesis (where the track length is zero for cascade-like events). Two distinct reconstruction methods are used for the ‘verification’ and ‘high stats’ samples respectively:

- 131 • SANTA/LEERA - geometric track reconstruction exploiting the hyperbolic light
132 emission profile on the string from the Cherenkov cone, followed by a cascade en-
133 ergy reconstruction
- 134 • RetroReco - a table-based full likelihood-based reconstruction utilising efficient ‘sink-
135 to-source’ reverse photon propagation tables

136 The main distinction between the two methods is that SANTA/LEERA is significantly
137 faster and simpler than RetroReco, but has an idealised event hypothesis and requires a
138 significant number of unscattered photons in the event, and as such has low efficiency ($\sim$
139 45%). RetroReco meanwhile can be used on all events and provides an improved treatment
140 of detector noise. RetroReco is somewhat comparable to the Pegleg reconstruction used
141 in the previous generation of oscillation analyses, and achieves similar resolutions but is
142 $\sim 5\times$ faster (reconstruction time is a major bottleneck of DeepCore analyses).

143 Both the ‘verification’ and ‘high stats’ samples have their own PID methods, but both
144 follow the same concept. The methods attempt to separate events with a track from those
145 without as a proxy for $\nu_{\mu,CC}$ event identification, utilising machine learning methods based
146 on BDT ensembles. The final level reconstructed track length and difference in fit quality
147 between cascade+track and cascade-only reconstructions are dominant features in both
148 cases.

149 1.5 Systematic uncertainties

150 The categories of systematic uncertainties considered in the analysis are:

- 151 • Detector and ice characterisation
- 152 • Neutrino flux modelling
- 153 • Atmospheric muon flux modelling
- 154 • Neutrino interaction modelling

155 The analyses use a floating normalisation with no prior (a ‘shape-only’ fit), and as such
156 are insensitive to overall shifts in neutrino rates from systematic uncertainties.

157 Detector/ice uncertainties are modelled by producing multiple simulation sets with
158 varying DOM efficiency, bulk ice scattering and absorption and hole ice parameters, and
159 implementing these parameters as nuisance parameters in the analyses. Hypersurfaces
160 fitted to these systematic sets are used to re-weight the nominal simulation sets during the
161 fit to reflect the nuisance parameters.

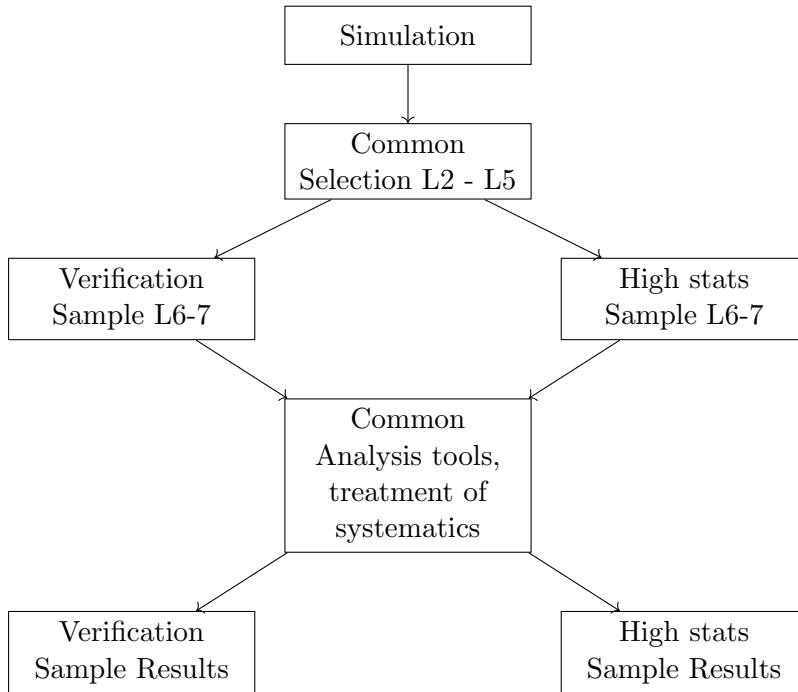
162 Neutrino flux uncertainty is modelled by using the MCEq tool to evaluate the impact
163 of varying the primary cosmic ray spectrum and hadronic production model parameters
164 (characterised by ‘Barr blocks’) on the predicted atmospheric neutrino flux. The varied

165 parameters are treated as nuisance parameters in the analyses. The atmospheric muon
 166 flux uncertainty is modelled using a normalisation and spectral index uncertainty.

167 The uncertainties in the quasi-elastic and resonant production neutrino interaction
 168 channels are modelled according to parameterisations provided by the GENIE event gen-
 169 erator. DIS uncertainties are modelled using a combination of varying GENIE parameters
 170 and by comparison for GENIE to accelerator neutrino interaction data.

171 1.6 Analyses

172 The overall analysis chain is as follows:



173 All analyses are performed by comparing a template MC histogram to a detector data
 174 histogram, where the template is re-weighted according to free parameters accounting for
 175 the physics and nuisance parameters. A minimizer is used to fit the parameters to best
 176 match the MC to data. This method is often referred to as ‘forward-folded parameter
 177 estimation’, and is the same as used in the previous generation of oscillation analyses.

178 A common analysis framework, ‘PISA’, is used for all analyses, and contains the MC
 179 template pipelines, systematic uncertainty implementations, and fitting tools.

180 The initial round of analyses (for which the initial review is conducted) are:

- 181 • ν_μ disappearance (‘verification’ sample)

182 • ν_μ disappearance (‘high stats’ sample)

183 • ν_τ appearance (‘high stats’ sample)

184 A good agreement between the results of the ‘verification’ and ‘high stats’ ν_μ disap-
185 pearance measurements will be an important test of the robustness of the analyses.

186 Subsequent analyses currently underway using the same sample are:

187 • Non-Standard Interactions (NSI)

188 • 3+1 sterile neutrinos

189 • Neutrino decoherence

190 • Neutrino Mass Ordering (NMO)

191 • Galactic center indirect Dark Matter search

2 Simulation

All simulation data can be found in `/data/sim/DeepCore/2018/pass2`.

The GCD file used for simulation is `/cvmfs/icecube.opensciencegrid.org/data/GCD/GeoCalibDetectorSt57531_PASS2_SPE_withScaledNoise.i3.gz`, which has the following hashes:

- `sha256sum 1fb25691cdd0de708fe8f1ef47a71df5ed5ef7b44766c517be26b841638ab929`
- `md5sum 68b2df1f328a3f0c6f3cd4bebec3605c`

2.1 General Pipeline

The low energy simulation chain produces artificial events in the DeepCore detector, and includes the following steps:

- **Step 1** Primary particle generation and propagation. Particles are created (using GENIE for neutrinos, MuonGun for muons) and propagated through the ice. Secondary particles and energy deposits are recorded. [[GENIE script](#), [MuonGun script](#)]
- **Step 2** Photon propagation. CLSim is used for photon propagation. [[script](#)]
- **Step 3** DOM response. This step turns photons into hits, using the given hole ice model and DOM Efficiency, applies Vuvuzela noise, PMT response, DOM Launcher, and TriggerSim. [[script](#)]
- **Level 1** Standard SimProd filtering scripts (pass 2 versions). This applies the filter masks and checks whether events passed any IceCube filters. This processing script can be accessed from any IceTray environment using:
`$I3_SRC/filterscripts/resources/scripts/SimulationFiltering.py` .
- **Level 2** Standard offline processing scripts (pass 2 versions). Hit cleaning and some fast reconstructions are performed at this stage. This processing script can be accessed from any IceTray environment using:
`$I3_SRC/filterscripts/resources/scripts/offlineL2/process.py` .

2.2 Noise simulation

The noise produced by the DOMs has two components:

Thermal noise comes from spontaneous emission of single electrons from the photocathode due to its thermal energy. The emission is uncorrelated among PMTs. Their number is drawn from a Poisson distribution and their times are uniformly distributed in the event's time window.

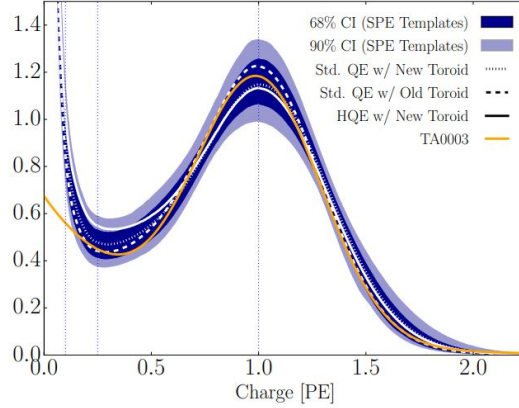


Figure 1: The inner (outer) dark blue region shows the 68% (90%) confidence interval defined by the measured spread in the extracted SPE charge templates of all DOMs in the detector. Superimposed are the average SPE charge templates for the variety of hardware configurations shown in white. The TA0003 distribution, for comparison, is shown in orange. All curves have been normalized such that the area above 0.25 PE is the same. Taken from the SPE template paper[3].

Non-thermal noise comes from radioactive decays in the glass making up the DOM pressure vessels and PMTs, which create bursts of several photons. The time of the decays is uniformly distributed in the event’s time window while the pulse times relative to each decay are sampled from a log-gaussian distribution.

The non-poissonian component has been studied in detail and the *vuvuzela* model that includes this was introduced to IceCube simulation in 2013-2014 [1]. The noise rates need to be updated if the basic behavior of the detector is modified, so the introduction of new SPE templates meant that the parameters of the model needed to be extracted once again.

Vuvuzela simulates noise as Monte Carlo Photoelectrons (MCPEs), and a charge needs to be assigned to them after they have been produced. The charge is sampled from the SPE distribution, which changed after *pass2*. The new distributions have a significant fraction of their area at low charge (see Fig. 1), so a substantial fraction of the photoelectrons from noise were being given a charge below the discriminator threshold, which was resulting in a lower effective noise rate. More details on the *pass2* SPE template calibration can be found in [2].

A complete re-fit of the model would be a very time consuming effort, since it requires special data to be taken as well as computationally intensive simulation. We decided instead to scale the individual components of the noise to recover the behavior that we had before the new SPE templates were introduced. The new parameters modify

- 241 • Thermal rate (TR)
- 242 • Decay rate (DR)
- 243 • Scintillation hits (SH)
- 244 • Scintillation mean (SM)
- 245 • Scintillation sigma (SS).

246 The new parameters introduce the following modifications. First, the thermal rate:

$$TR_{new} = TR_{old}/\eta. \quad (1)$$

247 To conserve the number of intervals with low time differences

$$SH_{new} = SH_{old}/\eta. \quad (2)$$

248 To conserve the total rate

$$DR_{new} = DR_{old} \frac{SH_{old} + 1}{SH_{old} + \eta}. \quad (3)$$

249 And a new timing:

$$SM_{new} = SM_{old} + \log_{10} \eta + \log_{10} \left(1 - (1 - \eta) \frac{SH_{old}}{SH_{old} + 1} \right) \quad (4)$$

250 The last term on the scintillation mean is not well motivated, but it had to be added
251 because without it the component with small time differences was observed to be wrong.

252 Figure 2 shows the disagreement created by the new SPE templates, and how the
253 scaling proposed above can mitigate that disagreement substantially. In Fig. 3 a similar
254 comparison can be made but now for noise-only events that actually trigger the detector,
255 using two low-level variables: the number of pulses in a cleaned series and the total charge
256 of these pulses.

257 The scaling constants to vuvuzela were developed in the oscillations WG first, but
258 they were shortly after discussed in the Calibration WG and also in collaboration-wide
259 calls. They have been accepted as the baseline model for any simulation using new SPE
260 templates.

261 2.2.1 Simulation of pure noise events

262 The initial trigger (**SMT3**) for this event selection is very sensitive to lower the energy
263 threshold. This has the side effect that pure noise events without any particles pass this
264 initial filtering step that have to be rejected at higher levels of the event selection. For
265 this reason, pure noise events have to be simulated in addition to muon and neutrino
266 interactions.

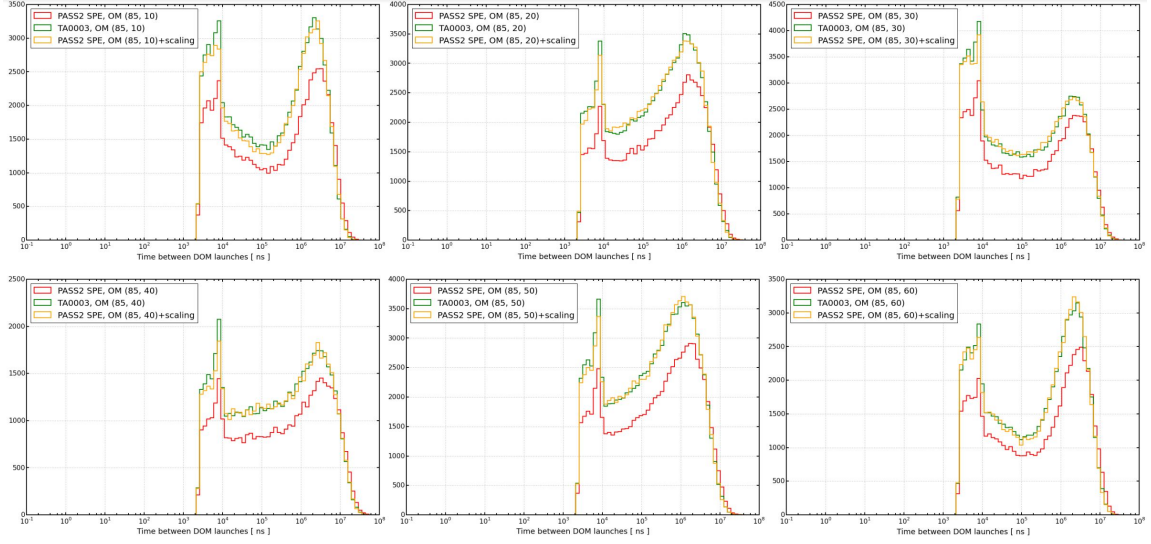


Figure 2: Behavior of the noise model as a function of time between DOM launches for pure noise simulation of six DOMs from string 85. Simulation before pass2 is shown in green (pass1), while simulation with pass2 SPE templates is shown in red (before scaling) and yellow (after scaling). The distributions agree well between pass1 and pass2 after scaling is applied.

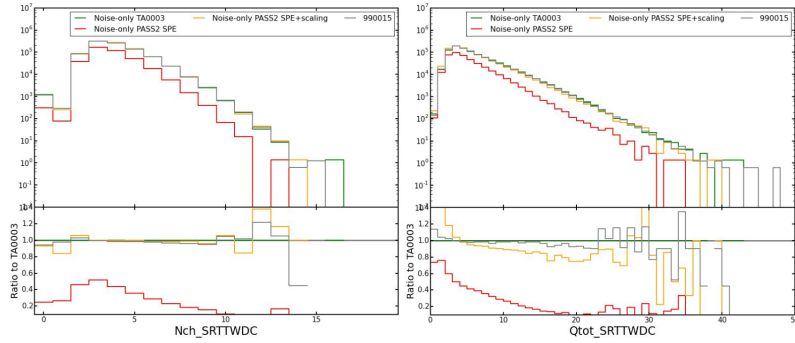


Figure 3: Comparison of events created by injecting pure noise in the entire detector with the original vuvuzela model, the modified SPE templates and the new scaling constants.

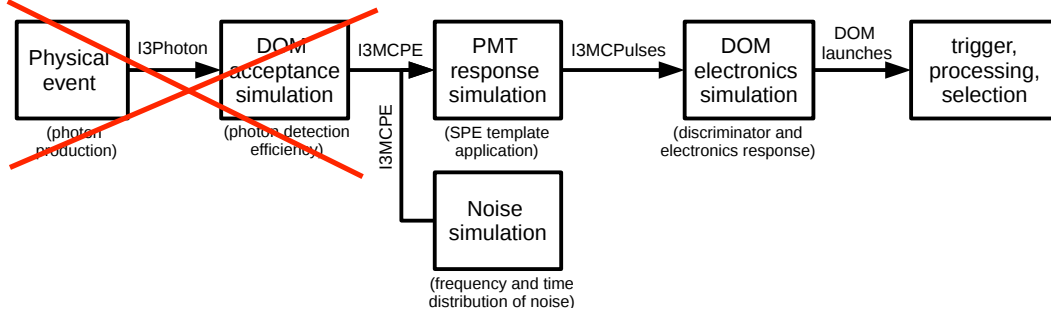


Figure 4: A simplified chart of the IceCube detector simulation chain for pure noise events. Physical event simulation steps are removed.

$$w = \frac{1 - 2800 \text{ [Hz]} \cdot 30 \text{ [} \mu\text{s]}}{3000 \cdot (0.1 \text{ [s]} - 20 \text{ [} \mu\text{s]})} \cdot \frac{1}{N_{files}} \approx \frac{0.003054}{N_{files}}$$

Muon rate and readout window
 Frames in file Frame duration Edge effects Number of files

Figure 5: Calculation of noise weight

267 Pure noise is simulated with the same simulation chain that is used in muon and
 268 neutrino simulations, except for the removal of the particle injection, photon propagation
 269 and DOM acceptance simulation steps as shown in figure 4.

270 The `vuvuzela` simulation injects MCPEs from thermal and non-thermal noise with
 271 parameters controlling the rates and time distributions that are fit for each DOM individ-
 272 ually. After this step, the simulation chain continues as usual with the simulation of the
 273 PMTs response to each injected MCPE. The weight for noise events follows the calculation
 274 shown in Figure 5.

275 2.3 GENIE simulation

276 Neutrino simulation is produced using the GENIE neutrino MC generator, interfaced with
 277 the IceCube software using `genie-icetray`.

278 The cross-sections for all neutrino flavors and interaction types are pre-calculated using
 279 stand alone (no IceCube software) GENIE version 2.12.8. They are calculated for energies
 280 from 0-10 TeV and their splines are saved to a xml file that is used during "Step 1" of the

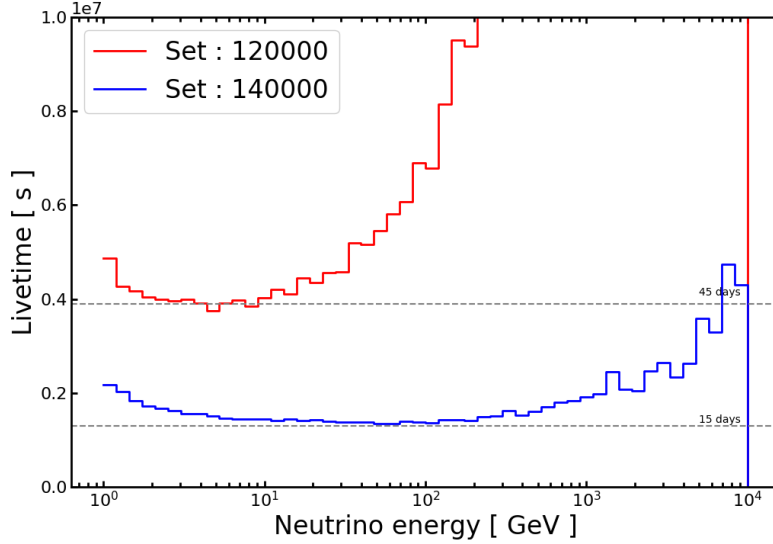


Figure 6: The live-time as a function of energy for a single ν_e (red line) and ν_μ (blue line) file. To obtain the total live-time one must multiply by the total number of files for that flavor.

281 general pipeline. Details on the models used by GENIE can be found in chapter 2 of the
 282 GENIE User Manual. [4]

283 All step 1 files were produced using the "step_1_genie.py" program and steered with
 284 the "genie_step1_job.sh" file from [this script](#). All processing was done on Compute Canada
 285 cluster Cedar so both of these files would need minor adjustment to be run elsewhere.

286 The I3MCTree is made with I3GENIEGenerator in the above mentioned scripts. A
 287 constant spectral index of $E^{-2.0}$ is used for all energy ranges. Neutrinos and anti-neutrinos
 288 are produced at a mixture of 70% to 30%, respectively. The total number of events/file
 289 and total number of files/flavor are summarized in the last two columns of Table 1. These
 290 numbers are chosen to allow for an energy-averaged total live-time of ≈ 70 years for each
 291 neutrino flavor ($\sim 10\times$ the detector data livetime)). The actual live-time as a function of
 292 energy is shown in Figure 6 (note that this depends on the assumed flux and oscillations,
 293 here the Honda 2014 flux is used).

294 The production of each flavor is split into multiple energy ranges from 1 GeV to 10
 295 TeV. This is done so each energy range can have its own generation volume; these volumes
 296 have been roughly optimized to be large enough to allow almost all events produced that
 297 could pass the DC L3 filter to be saved but not so large to be wasteful. A breakdown of the
 298 different energy ranges and their corresponding generation volumes are listed in Table1.

Table 1: Table of generation volumes used for GENIE MC production. The cylinder origin is (46.29, -34.88, -330.0) m (in the standard IceCube coordinate system) in all cases, which is centered on DeepCore.

Flavor	Energy (GeV)	Radius (m)	Length (m)	Events/file	Files
$\nu_e + \bar{\nu}_e$	1-4	250	500	450000	650
	4-12	250	500	100000	
	12-100	350	600	100000	
	100-10000	550	1000	57500	
$\nu_\mu + \bar{\nu}_\mu$	1-5	250	500	408000	1550
	5-80	400	900	440000	
	80-1000	450	1500	57500	
	1000-10000	550	1500	6700	
$\nu_\tau + \bar{\nu}_\tau$	1-4	250	500	1500000	350
	4-10	250	500	300000	
	10-50	350	600	375000	
	50-1000	450	800	200000	
	1000-10000	550	1500	26000	

PROPOSAL is used to propagate all μ^+/μ^- and uses "mediadef" found in "PROPOSAL/resources/mediadef" for the ice properties. A cylinder radius and length of 1200 and 1700 meters are used, respectively. Photon propagation is done in "step_2_clsims_setCrossE.py", also found [here](#) and steered with "genie_step2_job.sh". CLSim is utilized for this with the geometry defined in the GCD file `GeoCalibDetectorStatus_AVG_55697-57531_PASS2_SPE_withScaledNoise`. Hadronic (electromagnetic) showers with energy less than 30 GeV (100 MeV) are produced with GEANT4, while those with greater energy use the built in parametrizations. No DOM over-sizing is used.

The detector response is applied in `step_3_det_general_lowdt.py` of the "oscnex_scripts", and like the other steps is steered using "genie_step3_job.sh". This is a multi-step process running `I3CLSimMakeHitsFromPhotons` $\rightarrow$ Vuvuzela $\rightarrow$ PMTResponseSimulator $\rightarrow$ DOM-Launcher $\rightarrow$ I3DOMLaunchCleaning. For the baseline simulation, "as.flasher_p1_0.30_p2_1" is used for the hole-ice model. No cable shadowing is used in any of the sets. The default options are used for all of the other modules listed.

Finally, two specially made programs are run during the GENIE simulation chain. The first is "RemoveLatePhotons.py" run during step3. This is run to remove photons made by the decay of free neutrons. These decay on the time scale of minutes, causing very long time frames in files which we remove. The other is "convert_files.py" which rennumbers all events produced to have a unique Run/Event/SubEvent number. This fixes a bug in the Level1/2 scripts which causes all events to have the same event number. Both of these files

319 can be found [here](#).

320 2.4 MuonGun simulation

321 The atmospheric muon background is modeled using the MuonGun event generator [[doc-](#)
322 [umentation](#)]. The simulation chain is identical to GENIE simulation other than the event
323 generator itself.

324 The following settings and parameters are used as inputs (to the step1 processing) to
325 produce MuonGun simulation. The values used for these parameters can be found in Table
326 2.

- 327 • **Weighted Spectrum** Muons are *weighted* according to this flux spectrum.
- 328 • **Generated Spectrum** Muons are *generated* according to this spectrum.
- 329 • **Outer Surface** Muons are produced on a cylinder slightly larger than than the
330 IceCube volume, and are then propogated inwards.
- 331 • **Inner Surface** Muons are selected such that they must intersect an inner target
332 surface. The inner target volume is a cylinder surrounding DeepCore. This means
333 only muons passing through DeepCore will be simulated.
- 334 • **KDE Filter** Muons are selected such that their (unweighted) flux matches some
335 target distribution, using a kernel density estimator. This filter project can be found
336 [here](#). This is used to improve the efficiency of simulation, so that only events which
337 will be be important later on are simulated. A KDE tuned to oscNext L5 is used to
338 produce distributions optimised towards the analysis level.
- 339 • **Number of Events per File** This is the initial number of muons thrown. This is
340 *not* the number of events in the step1 files, because only the events that pass the
341 inner cylinder and KDE are kept.

Table 2: List of input variables and settings used to produce baseline MuonGun simulation.

Category	Setting	Value
Weighted Spectrum	Model	GaisserH4a_atmod12_SIBYLL
	Multiplicity	1
Generated Spectrum	Power Law Index	-3.0
	Power Law Offset	150
	Minimum Energy	150 GeV
	Maximum Energy	5 TeV
Outer Cylinder	Center (x,y,z)	(0,0,0) m
	Radius	800 m
	Length	1600 m
Inner Cylinder	Inner Cylinder Used	True
	Center (x,y,z)	(46.29,-34.88,-400.00) m
	Radius	180 m
	Length	400 m
KDE	KDE Used	True
	Target Distribution	oscNext_muongun_level5_v01.01_pass2 Set 139010, Files 000000-000999
	Floor Used	True
# of Events per File	NumEvents	50000000

Several iterations of simulation were produced. The sets directly used for OscNext development and analyses are described here. Details of the settings for each set can be found [here](#).

- **139008** A full-detector simulation set (e.g. no target inner volume) that can be used at low processing levels (L4 and below). Used for testing L3 and training the L4 classifier.
- **139010** DeepCore target volume used, but no KDE pre-scale applied. Used to develop L5.
- **139011 Official Baseline Set.**
- **130001-130004** DOM Efficiency Systematic Sets.
- **130100-130109, 130150-152** Hole Ice Systematic Sets.
- **130500-130503** Bulk Ice Systematic Sets.

2.5 Systematic Sets

Most simulation is produced according to our current best knowledge of how the detector operates, using values provided by the Calibration working group. However many of these aspects have uncertainty that could affect final results. To mitigate this, neutrino and muon simulation is also produced with varying choices of these parameters and propagated through the full event selection and reconstruction. This allows us to understand how uncertainties in the detector response could affect both the overall rates and the shapes of the distributions at analysis level.

2.5.1 DOM Efficiency

When a photon arrives at a DOM, there is some probability that it will produce a hit based on many factors, including the PMT quantum efficiency, the local ice scattering and absorption properties, the properties of the glass housing, etc. All of these factors are wrapped up into one parameter called DOM Efficiency that measures how efficient we are at collecting light. This factor is introduced as a scaling factor relative to the baseline set. During the detector response (step 3) of simulation, the probability of a photon being accepted is scaled up or down depending on the DOM Efficiency used for that set. Sets were produced at a DOM Efficiency of 0.9, 0.95, 1.05, and 1.1 (covering the region of $\pm 1\sigma$ of our prior on the DOM efficiency), while the baseline uses a relative DOM Efficiency of 1.0. The corresponding set numbers are listed in Table 3. Additional sets with more extreme values were added later.

Because this only relies on the detector response, the photon propagation is not repeated as a way to save on computation time. These systematic sets use baseline set files after photon propagation and then re-applies the altered detector response and then the full event selection chain.

Table 3: List of set numbers and the corresponding DOM Efficiency used. The x in the set number will be 2, 4, 6 for Genie nue, numu, or nutau respectively, or 3 for MuonGun. This is used so that the first two digits refer to the PDG encoding of the primary particle.

Set Number	DOM Efficiency Setting
1x0000 (nominal)	1.00
1x0001	0.90
1x0002	0.95
1x0003	1.05
1x0004	1.10
1x0005	0.50
1x0006	1.20
1x0007	0.70
1x0008	0.80

2.5.2 Hole Ice & Angular Acceptance

The refrozen ice columns surrounding each string have different optical properties than the rest of the ice in the detector, due to bubbles introduced during refreezing which causes increased scattering. Whether a nearby photon hits a DOM depends on these local properties and the geometry of the DOM. For example, a downgoing photon is not very likely to be accepted because it will not hit the PMT face unless it is scattered a lot. An upgoing photon is more like to hit the PMT face, however if it is perfectly upgoing, the bubble column can scatter it away. Many models have been proposed to describe these effects and how the likelihood of a photon being accepted is related to the direction at which it arrives.

A new unified model was developed that parameterizes many discrete sets into one model with two parameters, p_0 and p_1 (using principal component analysis). OscNext simulation was produced using this¹ unified model, whose implementation details can be found [here](#). [This presentation](#) provides an overview of the method, and [this presentation](#) discusses recommended parameter ranges.

The range of p_0 and p_1 values that cover most other established models are $-0.6 < p_0 < +0.6$ and $-0.1 < p_1 < +0.1$. Because of the strong dependence of these parameters on each other, sets were produced varying both parameters at the same time instead of one at a time. The values of p_0 and p_1 for the sets were chosen quasi-randomly (instead of purely randomly) in order to ensure uniform coverage. The physically allowed region of the parameters is significantly larger and spans $-2.0 < p_0 < +1$ and $-0.2 < p_1 < +0.2$.

¹version d6df05b15079004c7f95062fb8228580

399 The analysis allows the parameters to float freely within that region without a prior. A
400 few systematic sets are produced at more extreme values of the allowed parameter space
401 as shown in table 4, as well as *on-axis* sets where only p_0 or p_1 is varied (not both).

402 The angular acceptance curves are also normalized so that it doesn't affect the overall
403 rate at which photons are accepted, and instead only affects the direction from which
404 photons are preferentially accepted or rejected. The overall rate would be absorbed into
405 DOM Efficiency. For this reason, some sets were also produced in which both the angular
406 acceptance and the DOM efficiency were changed at the same time. The values of p_0 , p_1 ,
407 and DOM efficiency for each set are listed in Table 4, and the resulting angular acceptance
408 curves are shown in Figure 7.

Table 4: List of set numbers and the corresponding settings. The x in the set number will be 2, 4, 6 for Genie nue, numu, or nutau respectively, or 3 for MuonGun. This is used so that the first two digits refer to the PDG encoding of the primary particle. The last three sets (50, 51 & 52) are extreme points probing far (but allowed) regions of the phase space.

Set Number	DOM Efficiency	p0	p1
1x0000 (nominal)	1.00	+0.101569	-0.049344
1x0100	1.00	-0.0648	-0.1088
1x0101	1.00	-0.4839	-0.0171
1x0102	1.00	+0.2803	-0.0754
1x0103	1.00	+0.1122	+0.0035
1x0104	0.88	-0.0498	-0.0543
1x0105	0.93	-0.3729	+0.0349
1x0106	0.97	+0.2965	-0.0363
1x0107	1.03	+0.1244	-0.1132
1x0108	1.07	-0.0355	-0.0185
1x0109	1.12	-0.3142	-0.0773
1x0150	1.00	-1.3	0.15
1x0151	1.00	-1	-0.1
1x0152	1.00	0.5	0.15
1x0300	1.00	-1	-0.049344
1x0301	1.00	-0.5	-0.049344
1x0302	1.00	+0.1	-0.049344
1x0303	1.00	+0.3	-0.049344
1x0305	1.00	+0.101569	-0.15
1x0306	1.00	+0.101569	-0.1
1x0307	1.00	+0.101569	+0
1x0308	1.00	+0.101569	+0.05
1x0309	1.00	+0.101569	+0.1
1x0310	1.00	+0.101569	+0.15
1x0311	1.00	-0.2	-0.049344

409 Note that the baseline simulation set is produced with the older "MSU" holeice model
410 with the parameters dima=0.3 + MSU p2=-1. This corresponds to p0 = 0.101569, p1 =
411 -0.049344 in the unified model.

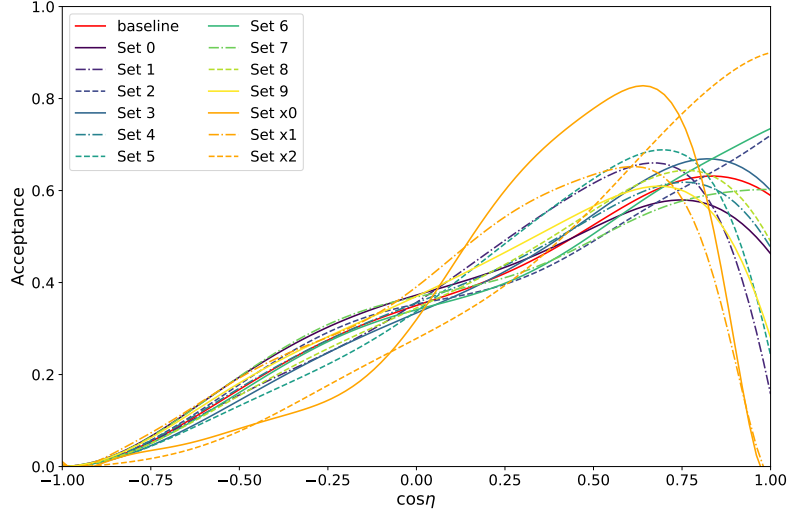


Figure 7: Angular acceptance curves corresponding the p_0 , p_1 parameters used for all systematics sets listed in Table 4. Sets 0 to 9 correspond to oscNext MC Set numbers 1x0100 to 1x0109, and sets x0 to x2 correspond to MC Sets 1x0150 to 1x0152, respectively. η is the photon incidence angle relative to the DOM, with $\cos \eta = 1$ being a photon travelling completely upwards into the downwards-pointing DOM.

412 2.5.3 Bulk Ice

413 In total 12 different bulk ice sets are simulated to take into account possible mis-modeling
 414 of the bulk ice absorption and scattering properties. The default model of **SPICE 3.2.1**
 415 is used but four systematic sets vary the scattering and absorption measured in the ice by
 416 $\pm 5\%$, corresponding to $\pm 1\sigma$ of our prior on these parameters. Additionally, a number of on-
 417 axis sets are produced with more extreme perturbations. For all these sets a DOM efficiency
 418 of 1.0 is used and the default hole ice model **as.flasher_p1_0.30_p2_-1** is applied. Table 5
 419 lists the exact bulk ice models used and their corresponding systematic set numbers. All
 420 sets are produced using the same step2 and step3 production and steering scripts listed in
 421 Section 2.3.

Table 5: List of set numbers and the corresponding bulk ice model used for the four different bulk ice systematic sets produced.

Set Number	Scat	Abs	Ice Model Used
1x0500	+5%	+5%	spice_3.2.1/err_s+.05_a+.05
1x0501	+5%	-5%	spice_3.2.1/err_s+.05_a-.05
1x0502	-5%	+5%	spice_3.2.1/err_s-.05_a+.05
1x0503	-5%	-5%	spice_3.2.1/err_s-.05_a-.05
1x0504	-	+10%	spice_3.2.1/err_s+.00_a+.10
1x0505	-	-10%	spice_3.2.1/err_s+.00_a-.10
1x0506	+10%	-	spice_3.2.1/err_s+.10_a+.00
1x0507	-10%	-	spice_3.2.1/err_s-.10_a+.00
1x0508	-	+30%	spice_3.2.1/err_s+.00_a+.30
1x0509	-	-30%	spice_3.2.1/err_s+.00_a-.30
1x0510	+30%	-	spice_3.2.1/err_s+.30_a+.00
1x0511	-30%	-	spice_3.2.1/err_s-.30_a+.00
1x0512	-	+20%	spice_3.2.1/err_s+.00_a+.20
1x0513	-	-20%	spice_3.2.1/err_s+.00_a-.20
1x0514	+20%	-	spice_3.2.1/err_s+.20_a+.00
1x0515	-20%	-	spice_3.2.1/err_s-.20_a+.00
1x0516	-	+5%	spice_3.2.1/err_s+.00_a+.05
1x0517	-	-5%	spice_3.2.1/err_s+.00_a-.05
1x0518	+5%	-	spice_3.2.1/err_s+.05_a+.00
1x0519	-5%	-	spice_3.2.1/err_s-.05_a+.00

2.5.4 Birefringence

In the time since oscNext MC production began, birefringence (BFR) was identified as a significant effect in photon propagation within IceCube. The impact of BFR on the final level oscNext sample was found to be significant (see [here](#)), and a number of MC sets with `spice_bfr-v2` were produced (including systematic sets with variations to the other detector systematic parameters), and are listed in Table 6.

Table 6: List of GENIE MC sets numbers that use BFRv2 ice. The script defining these sets is [here](#).

Set	Ice	DOM ϵ	HI p0	HI p1	Abs.	Scat.
1122	spice_bfr-v2	-	+0.101569	-0.049344	-	-
1124	spice_bfr-v2	+10%	-	-	-	-
1125	spice_bfr-v2	-10%	-	-	-	-
1126	spice_bfr-v2	-	-0.2	-	-	-
1127	spice_bfr-v2	-	+0.3	-	-	-
1128	spice_bfr-v2	-	-	-0.1	-	-
1129	spice_bfr-v2	-	-	0	-	-
1130	spice_bfr-v2	-	-	-	+5%	-
1131	spice_bfr-v2	-	-	-	-5%	-
1132	spice_bfr-v2	-	-	-	-	+5%
1133	spice_bfr-v2	-	-	-	-	-5%

3 General event selection

All event selection data can be found in `/data/ana/LE/oscNext/pass2`.

3.1 Overview

The main goal of the event selection in DeepCore is to achieve a neutrino-dominated event sample by rejecting a prevailing background from atmospheric muons and detector noise. The selection, described in this document, is split into different levels. The first levels typically use simple and fast algorithms to reduce the background rate to a level where more sophisticated and computationally intensive algorithms can be efficiently applied. The following sections cover the basic event selection and describe the steps to achieve a neutrino-dominated data sample.

3.2 Definitions

The following definitions are used throughout the event sample:

- *uncleaned pulses* are reco pulses before hit cleaning (frame object is `SplitInIcePulses`).
- *cleaned pulses* are reco pulses after hit cleaning (frame object is `SRTTWOofflinePulsesDC`).
- *pass2* refers to detector data (re-)processed using the 2017 season configuration (software versions, online charge calibration). The software used is `/cvmfs/icecube.opensciencegrid.org/v3.1.1/metaprojects/icerec/V05-02-04`. See [wiki](#) for more details.

3.3 Detector run selection

The detector data selected for analysis consists of runs featured in the *pass2 goodrunlist* database for every IC86 seasons (2011 to 2019). The run files at Level2 can be found on the datastore (each has a dedicated GCD file):

- `/data/exp/IceCube/YYYY/filtered/level2pass2a/` for 2011-2016
- `/data/exp/IceCube/YYYY/filtered/level2/` for 2017-20

The runs used in the event selection and analyses (as well as simulation datasets) are steered by the following [run database](#) (read by the `RunDatabase` class) good run conditions. The following conditions must be met for a run to be used (defined in the `"run_conditions"` entry in the run database), which are chosen based on the plots found [here](#):

- Marked as `good_i3` in I3Live

- 458 • Must have ≥ 2 hours of livetime
- 459 • Must have all 86 strings active
- 460 • Must have ≥ 5035 active DOMs in the run

461 The following runs have also been excluded after manual inspection:

- 462 • All runs prior to 118550 (2011) - excludes unstable runs from early 2011 (see section
463 E.2)
- 464 • Run 122205 - missing in I3Live
- 465 • Run 122206 - missing in I3Live
- 466 • Run 122207 - missing in I3Live
- 467 • Run 129028 - Was added late (Dec 19) to the 2016 GRL. 15 subruns fail with a
468 suspicious segfault.

469 Combined, these conditions result in $\approx 95\%$ of the total operating time of the detector
470 from run 118550 onwards being used in oscNext.

471 Additionally, a small number of sub-run files that are marked as good are failing the
472 Level 3 processing due to missing/corrupt frame objects in the L2 files. The full list of
473 these subruns is recorded in JSON files that are listed [here](#). These are excluded.

474 3.4 Level 2 (DeepCore Filter)

475 The selection begins at the normal Level 2 selection created for the entire IceCube collab-
476 oration. At Level 2, the raw waveforms acquired by the DAQ have already been calibrated
477 and reconstructed with wavedeform; the online filters, which select a subset of triggered
478 events that have a physics potential from each working group, send via satellite the cal-
479 ibrated data to the North, in files named *PFFilt*. The Level2 scripts are then run on
480 these files, to run more complicated filters which require computationally more expensive
481 reconstructions that cannot be run at Pole.

482 In the case of oscNext (and more generally within the low-energy working group),
483 the only filter which matters for the event selection is the **DeepCore Filter**, which has
484 existed in its current form since 2013. The filter looks for triggered events with more than
485 3 single multiplicity trigger (SMT3). It then applies a dedicated SRT cleaning on the
486 SplitUncleanedInIcePulses, using a slightly different configuration than the collaboration-
487 wide SRT cleaning (it namely keeps all HLC hits rather than just the core ones). The
488 SRT-cleaned pulse series is then split into a Deepcore fiducial and a DeepCore Veto series.
489 From that point, the main veto algorithm is run: first, the center-of gravity (COG) of the
490 hits inside the fiducial volume is calculated, then a relative velocity is derived between each

hit of the veto region and the COG’s position/time vertex. If the derived speed of any veto hit is contained within $[0.25, 0.4]$ m/ns, the hit is discarded on the basis that the veto hit could be causally related to a muon crossing the detector.

3.5 Level 3

The data passing DC Level2 filter is then processed by the DeepCore Level3 [\[script\]](#). The objective of Level3 is to use simple, fast cuts to remove regions of data-MC disagreement that result from event types for which we do not have MC, for example muon bundles, coincident events and SLOP triggers, and any other observed discrepancies caused by MC mismodelling.

This level was updated for the oscNext to avoid dependence on the PMT charge, and thus make it less sub-sceptible to SPE template mis-modelling (this philosophy has been used throughout the processing levels). The 2012-version of the Level 3 has a strong dependence on the charge of PMT pulses, resulting in significant change of the passing event rate with Pass2 calibration of the IceCube detector. In the version described here, all the charge-based variables were replaced with the closest analogs based on the number of DOMs registering light.

3.5.1 Variable description

DeepCore Level 3 cuts on a set of simple variables that are fast to calculate, with the aim of removing ‘easy’ muon and noise events, and improving data-MC agreement to a level that machine learning techniques can be applied in subsequent processing levels. The explanation of the variables is given in the following paragraphs. The variables are stored in the map frame object `IC2018_LE_L3_Vars`.

Broadly speaking, muons are identified by looking for light in the outer/upper (veto) regions of the detector. Noise is identified by looking at the times or numbers of hits. Coincident events are identified as longer in duration.

The bottom 50 DOMs on DeepCore strings and bottom 21 DOMs on IceCube strings further than 3 string from the detector edge are considered *fiducial DOMs* for DeepCore Level3, while other modules are *veto DOMs*. String and DOM numbers belonging to the fiducial volume are summarized in the Table 7

Table 7: A list of DeepCore Level3 fiducial strings and DOM numbers.

String type	String number	DOM numbers
DeepCore	79–86	11–60
IceCube	25–27, 34–37, 44–47, 54	39–60

520 The variables calculated are:

- 521 • **Number of cleaned DOMs (NchCleaned)**: A number of hit DOMs in the cleaned pulse
522 series.
- 523 • **NoiseEngine**: DeepCore Level3 uses the **NoiseEngine** algorithm without charge infor-
524 mation.
- 525 • **MicroCount hits**: The reconstructed pulse series withing $[-4 \mu s, +5 \mu s]$ from Deep-
526 Core are scanned with a 300 ns sliding time window. The algorithm saves a number
527 and charge of the time window with the maximal observed number of hits.
- 528 • **NAbove200**: A number of observed pulses in DOMs with $Z > -200m$ before the
529 DeepCore trigger time.
- 530 • **VertexGuess Z**: Z-coordinate of the first pulse in the cleaned pulse series.
- 531 • **Fiducial hits (DCFiducialHits)**: A number of hit fiducial DOMs in the cleaned pulse
532 series.
- 533 • **Veto/fiducial hit ratio**: Ratio of hit numbers in veto and fiducial modules in
534 the cleaned pulses.
- 535 • **Causal Veto Hits**: A number of veto hits found by the **DeepCore filter** (run at level
536 2). the time considered for each hit is the time of the first pulse in a DOM's pulse
537 series.
- 538 • **C2HR6**: A fraction of hit DOMs within the first 600 ns with the first 2 hit DOMs
539 ignored.
- 540 • **Uncleaned time length**: A time difference between the first and last pulse in the
541 uncleaned pulse series.
- 542 • **Cleaned time length**: A time difference between the first and last pulse in the
543 cleaned pulse series
- 544 • **RT Veto**: A number of hits found by the RTVeto algorithm. A different cut value is
545 selected for different ranges of hit fiducial DOM numbers as described in the table
546 below:

DC fiducial DOMs	RTVeto cut
$N_{fid} < 75$	RTVeto250Hits < 4.0
$75 \leq N_{fid} < 100$	RTVeto250Hits < 5.0
$N_{fid} \geq 100$	no cut

547

Table 8: A list of variables, their names in the output dictionary and cuts used for the DeepCore Level 3 (version 2018).

Variable	Fig.	Name in dictionary	Targets	Cut value
Cleaned DOM number	Fig. 8	NchCleaned	Noise	≥ 6
NoiseEngine	Fig. 8	NoiseEngineNoCharge_bool	Noise	<code>==True</code>
MicroCount hits	Fig. 8	STW9000_DTW300Hits	Noise	> 2
Fiducial hits	Fig. 8	DCFiducialHits	Noise	> 2
NAbove200	Fig. 9	NAbove200Hits	Muons	< 10
VertexGuess Z	Fig. 9	VertexGuessZ	Muons	< -120 m
Causal Veto Hits	Fig. 9	CausalVetoHits	Muons	< 7
Veto/fiducial hit ratio	Fig. 9	VetoFiducialRatioHits	Muons	< 1.5
C2HR6	Fig. 10	C2HR6	Muons	> 0.37
RT Veto	Fig. 10	RTVetoCutHit	Muons	<code>== True</code>
Uncleaned time length	Fig. 10	UncleanedFullTimeLength	Coincident events	< 13000 ns
Cleaned time length	Fig. 10	CleanedFullTimeLength	Coincident events	< 5000 ns

3.5.2 Cuts and control distributions

Variables described in the previous section are used to select the events passing the DeepCore Level 3 criteria. A list of variables, their names in the data files, control distributions and cut values are summarized in Table 8. Comparisons of these distributions in data and MC are shown in Figures 8, 9 and 10. As previously mentioned, good data-MC is not expected in these distributions pre-cut.

The performance of Level 3 cuts is tested using neutrino GENIE simulations for signal (sets 129002, 149002, 169002), MuonGun simulations for atmospheric muon background (set 139008) and Vuvuzela pure noise simulations (set 888003). Table 9 lists the rates and passing fractions of the different simulation sets. The rates are compared to runs 124750, 124900, 125400, 125700, 125900, 126250 of season 2014 (pass2 data processing).

Currently available MuonGun simulations do not include muon bundles and, in general, poorly describe bright muon events. However, such events are easier to reject. Example of such disagreements can be seen in Figure 8. Most of the disagreements are located in the muon-dominated regions, which are rejected by Level 3 cuts. Therefore, MuonGun simulations are expected to have a decent agreement at the later stages of the selection.

A small fraction of events passing the L3 filter are encompassed inside very long, **SLOP** triggers designed to search for slow moving monopoles. Because of the very large time window of these triggers (several ms), it can happen that a series of pulses within that trigger satisfy the DeepCore criteria. However, these events are not simulated properly, and are thus cut. It was shown in previous event selections that they can produce mismatch

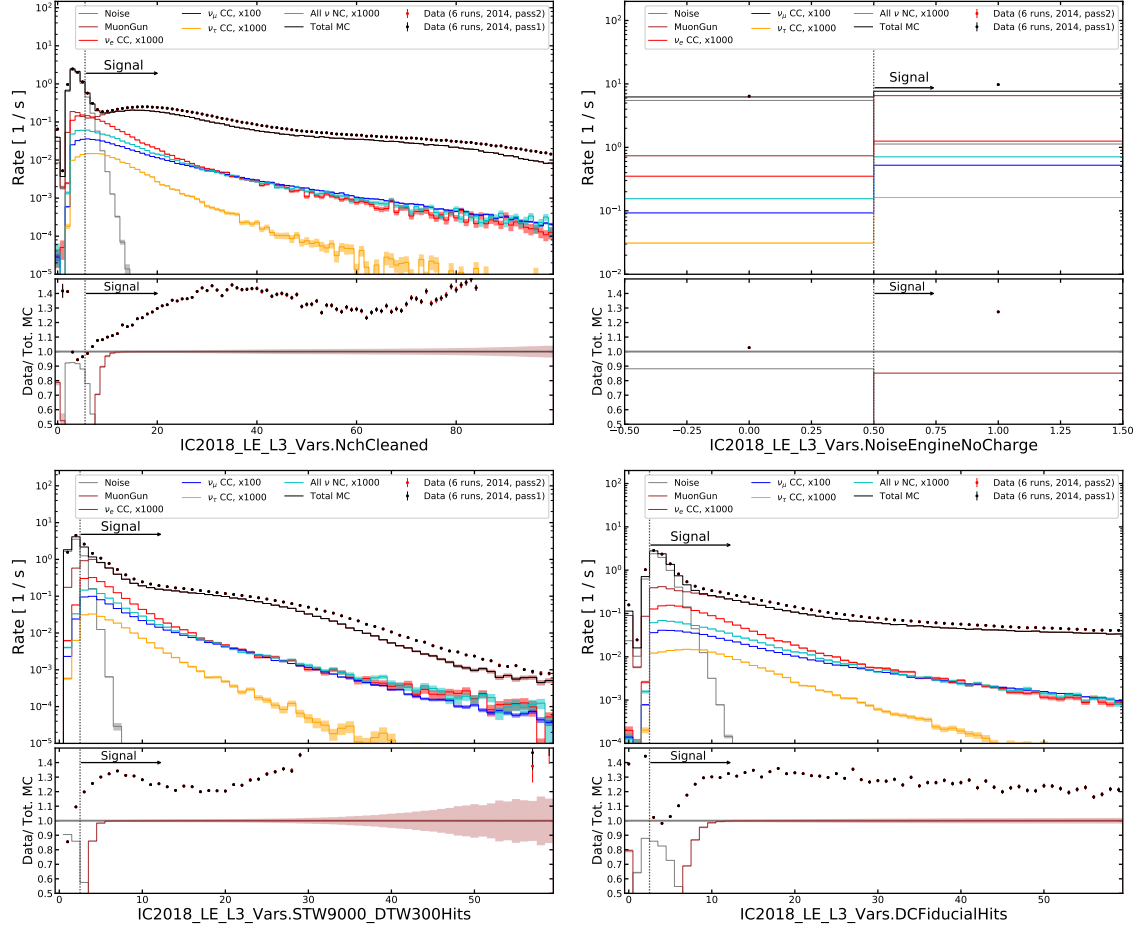


Figure 8: (Top left) Distribution of the number of hit DOMs in the cleaned pulse series; (top right) Distribution of the NoiseEngine output; (bottom left) MicroHits distribution; (bottom right) Number of hit fiducial DOMs

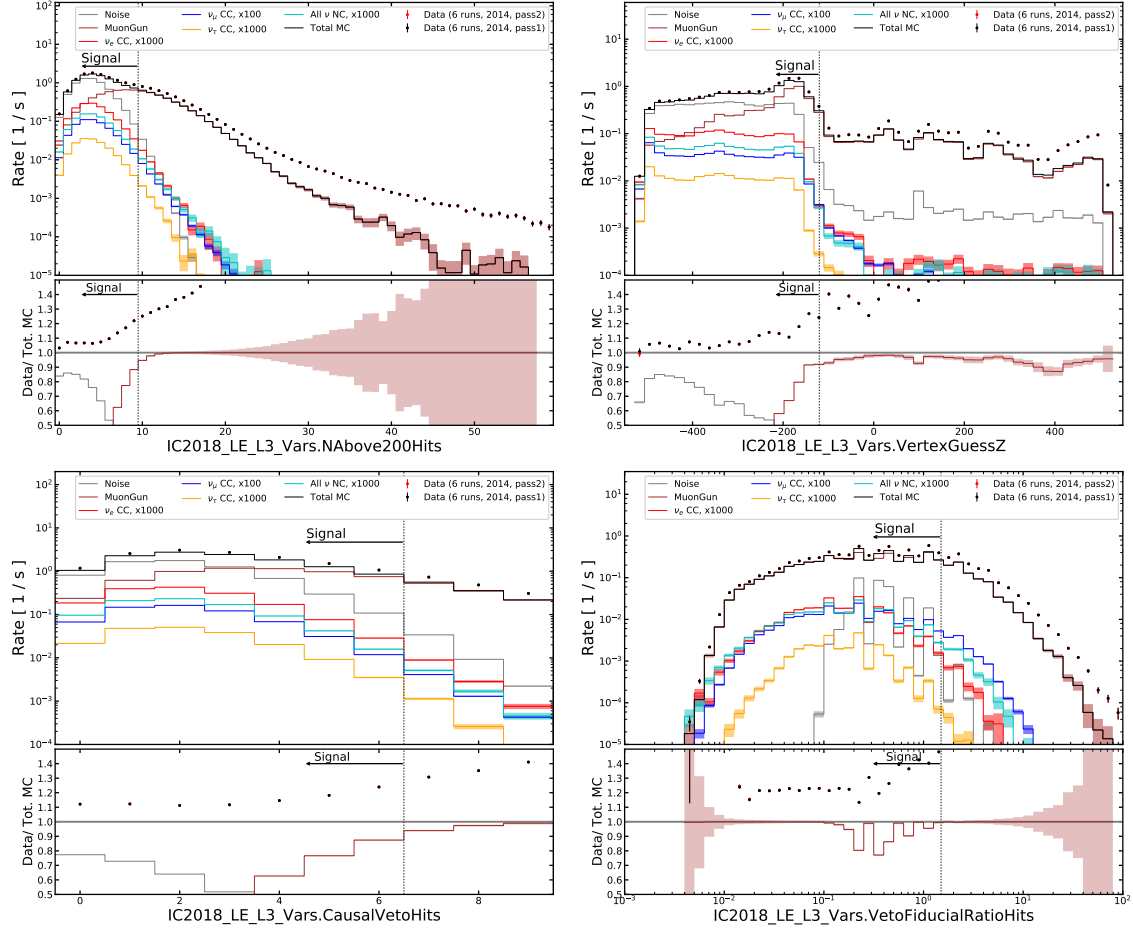


Figure 9: (Top left) NAbove200 distribution. ; (top right) VertexGuessZ distribution; (bottom left) CausalVetoHits distribution; (bottom right) Veto/fiducial hit ratio.

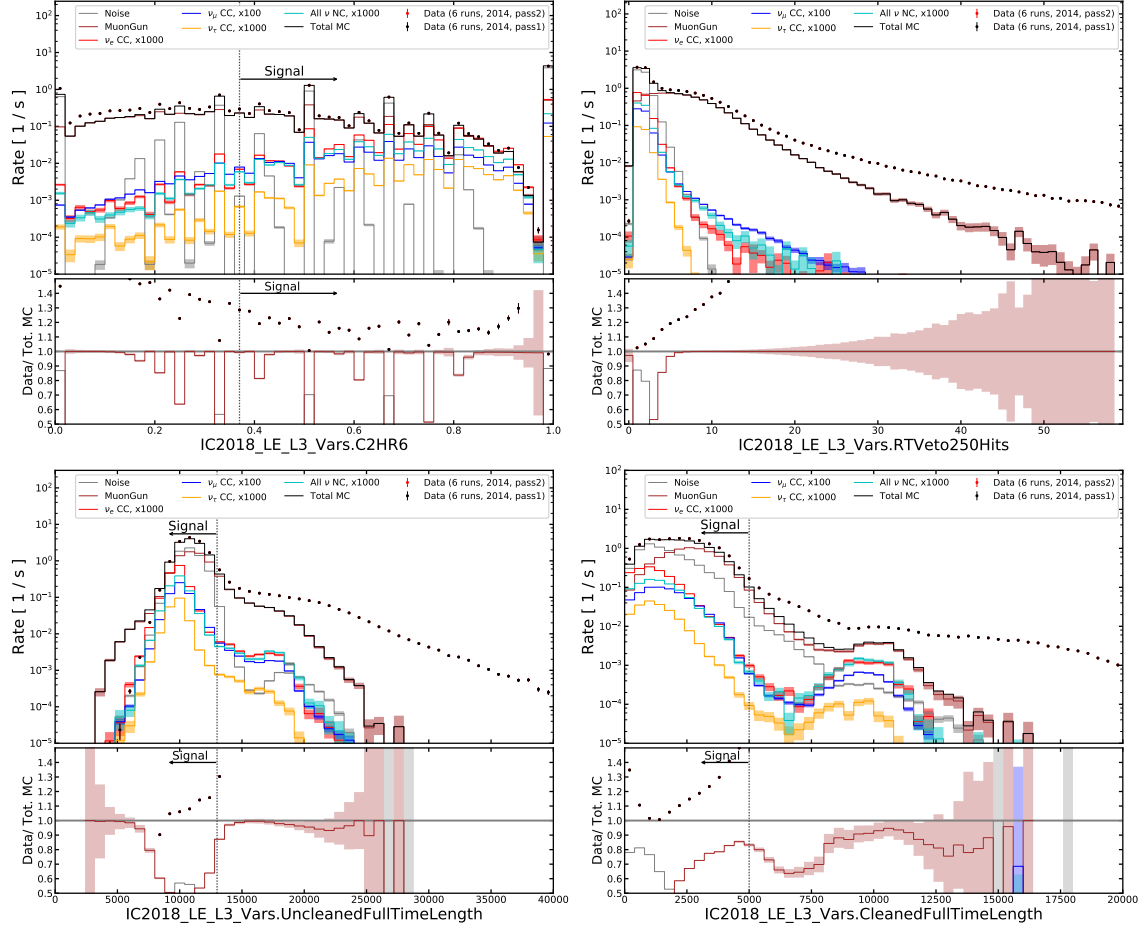


Figure 10: (Top left) C2HR600 distribution; (top right) RTVetoHits distribution, the cut criteria are not shown for this distribution, since different cuts are applied for different ranges of number of hit fiducial DOMs; (bottom left) Duration of the uncleaned pulse series; (bottom right) Duration of the cleaned pulse series. A clear data excess at the long events indicates absence of the coincident event simulations.

Table 9: Event rates at DeepCore Level2 filter and at new Level3 selection. Efficiency of the L3 cuts (relative to L2) is also shown.

	Rate@DC L2	Rate@L3	Efficiency
GENIE, ν_e CC	1.61 mHz	0.95 mHz	58%
GENIE, ν_μ CC	6.16 mHz	3.77 mHz	61%
GENIE, ν_τ CC	0.193 mHz	0.129 mHz	67%
GENIE, ν NC (all)	0.86 mHz	0.53 mHz	62%
MuonGun, atm. μ	7273 mHz	505 mHz	6.9%
Vuvuzela, pure noise	6621 mHz	36.6 mHz	0.6%
Total MC	13903 mHz	547 mHz	3.9%
Data (2014, pass2)	16168 mHz	582 mHz	3.6%

in the distribution of reconstructed goodness-of-fit values.

Finally, a cut on events containing flaring DOMs is applied, identified by the `LIDerrata` frame object. Flaring DOMs are cases where a very large amount of charge is temporarily seen on a single DOM, and is expected to be caused by effects in the DOM itself rather than any physics event.

3.5.3 Event rates and performance

The selection criteria of the Level 3 successfully removes approximately 95 % of background muon events and over 99 % of pure noise events, relative to level 2. At the same time, it keeps over 60 % of atmospheric neutrino events. The selection efficiency of different neutrino flavours can be found in Table 9.

A passing fraction of different neutrino flavours as a function of energy and zenith angles can be found in Fig. 11. The selection efficiency for neutrinos between 10 and 100 GeV is approximately 70 %. Events with lower energies are lost due to the rejection of pure noise events and requirement of the minimal light amount in the detector. Higher energy events (especially muon neutrinos) are lost due to cuts on light in the veto region.

3.6 Level 4

After the Level 3 selection, the event rate is under 1 Hz. Crucially, the data-MC agreement of the sample is much improved, allowing machine learning methods to be applied for efficient background rejection. The Level 4 event selection `[script]` makes use of machine learning classification methods to separate for muon and noise backgrounds from signal events.

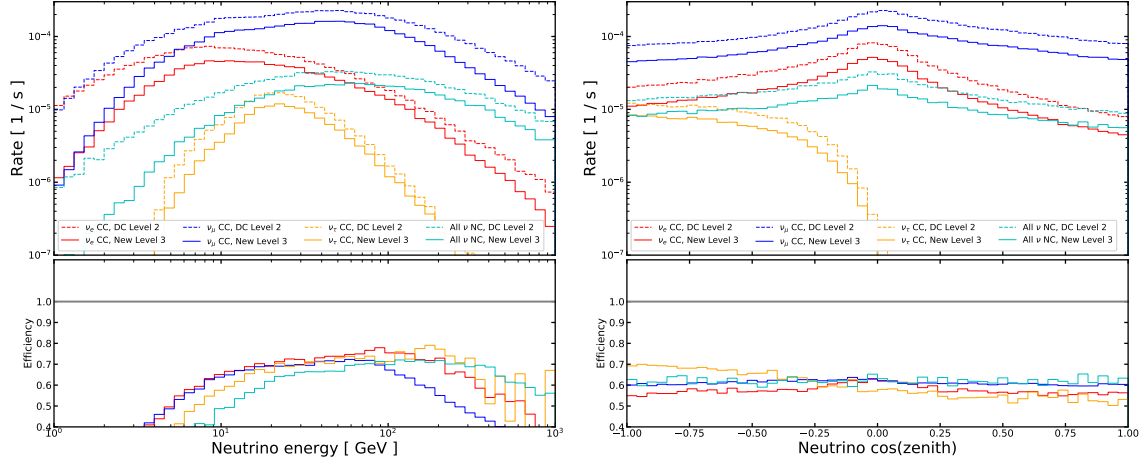


Figure 11: Passing fraction of neutrinos of different flavours as a function of neutrino energy (left) or cosine of the neutrino zenith angle (right).

3.6.1 Classifiers overview

Two classifiers are trained, one for identifying noise events and the second for identifying muon events. In both cases, the **LightGBM** algorithm is used for the classifier, which uses gradient boosting to train ensembles of Boosted Decision Trees (BDTs). The similar XGBoost and CatBoost algorithms were also tested, and produced comparable but slightly inferior results. All classifiers are integrated with IceCube software using the **Classifier** class.

The input data is split into ‘train’ and ‘test’ used to train and subsequently verify respectively the machine learning models. Good train-test agreement is seen, indicating that overtraining is not significant, and the train samples are not discarded in the analysis. The settings (‘hyperparameters’) used for the level 4 classifiers are listed in Table 10. These settings are chosen to minimize overtraining.

~40 variables were tested in total as inputs to the classifiers, with the selected variables being those showing good data-MC agreement and strong ‘feature importance’ (classification power). The number of variables passed to classifiers was minimised to limit the model complexity and hence reduce the possibility of overtraining.

Some pre-processing is performed on the input data to improve the classifier performance. The event weights were manually re-scaled to ‘balance’ the samples, e.g. to give equal total weighted numbers for signal and background. In addition, all the weights were re-scaled to be in the range between 0 and 1 to avoid accumulation of numerical errors.

Table 10: Settings given to the LightGBM algorithm for training of noise and muon rejection algorithms. The main goal of selected settings is to avoid overtraining with an available amount of the detector simulations.

Parameter	Noise BDT	Muon BDT
is_unbalanced	False	
max_depth	6	
num_leaves	25	
max_bin	32	
max_bin	32	
min_data_in_leaf	500	
feature_fraction	0.8	0.7
lambda_l1	2.0	
lambda_l2	1.0	
min_gain_to_split	2.0	

3.6.2 Classifier for noise rejection

The target for the noise classifier is to provide a robust algorithm based on only few variables to separate noise from neutrino events. The classifier was trained on Vuvuzela noise-only simulations (set 888003) and GENIE neutrino simulations (sets 1X9002 - an earlier version of the 1X0000 sets). The training fraction was selected at 33.3%, while 66.7% of the simulations are used as verification sample.

Table 11 lists the 5 input variables used in the noise classifier, with the variable distributions for data and MC shown in Figures 12 and 13. The ‘feature importance’ of all classifier input variables is shown in Figure 14, showing that all variable are contributing significantly.

The noise classifier achieves a strong separation of noise and neutrino events, shown in Figure 15. The noise and neutrino rates as a function of cut value on the classifier prediction are shown in Figure 16, showing that the noise rate can be reduced by orders of magnitude (to below the neutrino rate) whilst maintaining good neutrino rates. Some overtraining is observed in the noise component, but ultimately noise is reduced to negligible levels by the end of the selection and thus any impact of this overtaining on the sample is negligible.

The classifier prediction is written to the frames as L4_NoiseClassifier_ProbNu, and its distributions for data and MC are shown in Figure 17. Only events with the neutrino probability score above 0.7 are accepted in the sample. This reduces the rate of noise-only events by approximately 2 orders of magnitude from about 36.6 mHz to less than 0.3 mHz, while keeping $\sim 96\%$ of all neutrino events.

Table 11: Level 4 noise classifier input variables.

Variable	Description
<code>IC2018_LE_L3_Vars.NchCleaned</code>	See description in section 3.5.1
<code>L4_micro_count.STW_m3500p4000_DTW200</code>	Sort with the cleaned pulse series. Look at pulses occurring within $[-3.5 \mu\text{s}, +4 \mu\text{s}]$ from the trigger time. Slide a time window of 200 ns that maximizes the number of triggered DOMs in it. Get the number of triggered DOMs in that sliding time window
<code>L4_iLineFit.speed</code>	Speed fitted by the improved LineFit algorithm
<code>L4_fill_ratio.fill_ratio_from_mean</code>	Measure of the geometrical spread of the hits about some vertex (details here)
<code>IC2018_LE_L3_Vars.FullTimeLengthRatio</code>	Measure the total duration of an event in both the cleaned and uncleaned pulse series (duration = max_pulse_time - min_pulse_time). Calculate the ratio between the two.

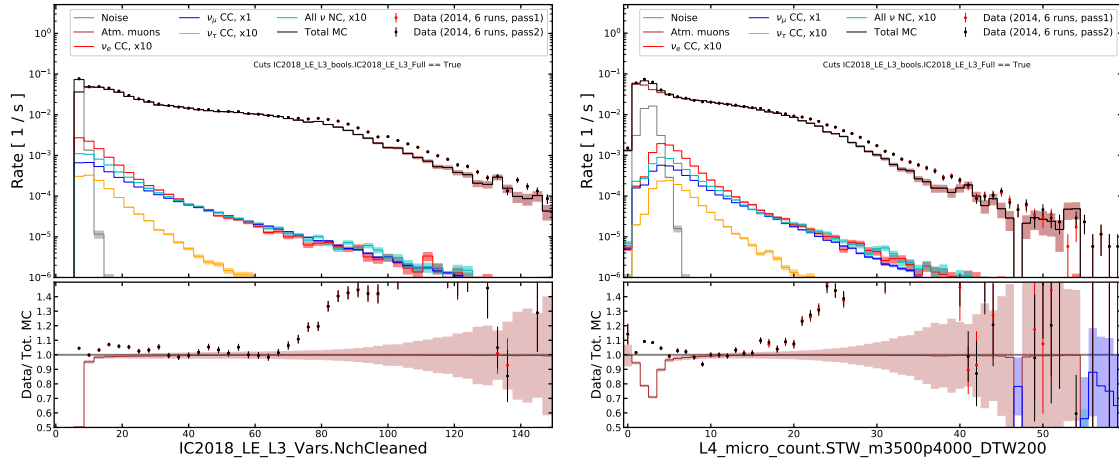


Figure 12: `IC2018_LE_L3_Vars.NchCleaned` and `L4_micro_count.STW_m3500p4000_DTW200` distributions for data and MC.

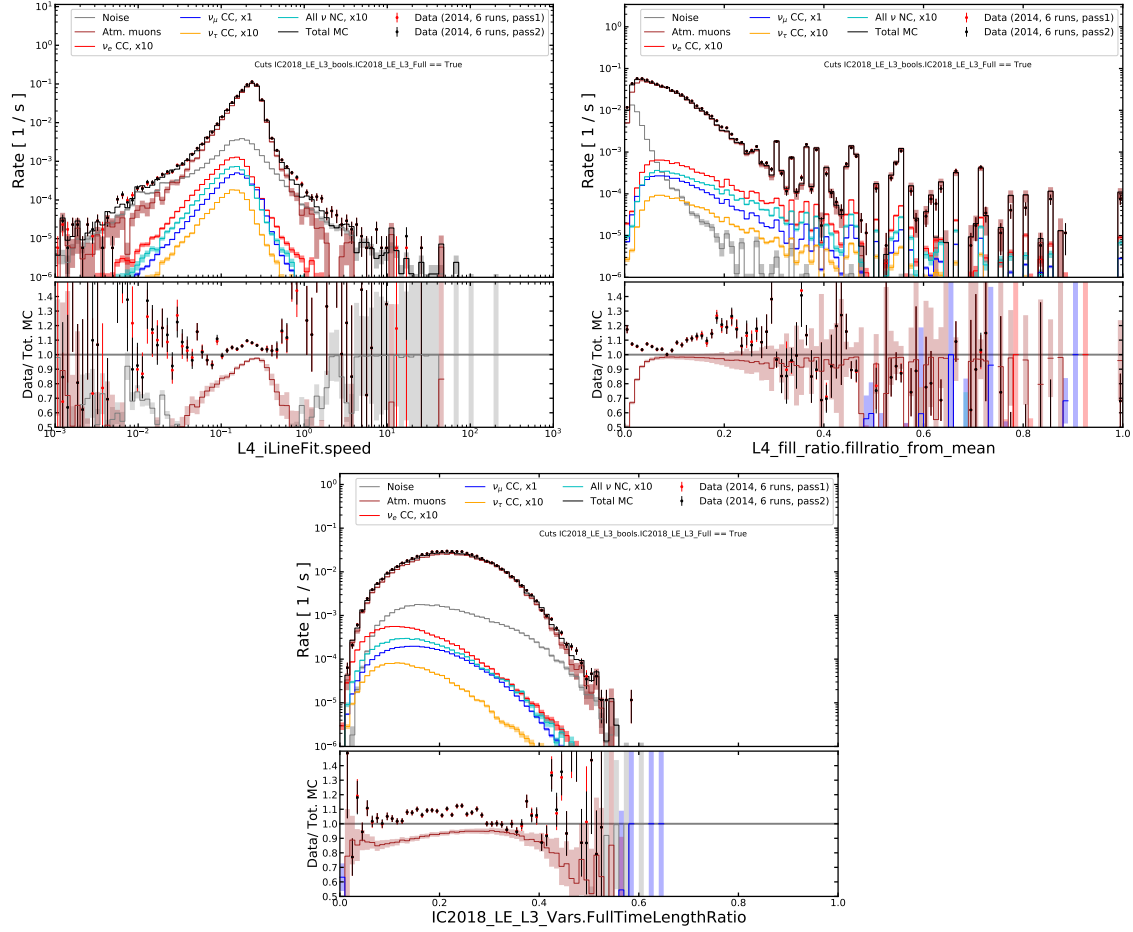


Figure 13: `L4_iLineFit.speed`, `L4_fill_ratio.fillratio_from_mean` and `IC2018_LE_L3_Vars.FullTimeLengthRatio` distributions for data and MC.

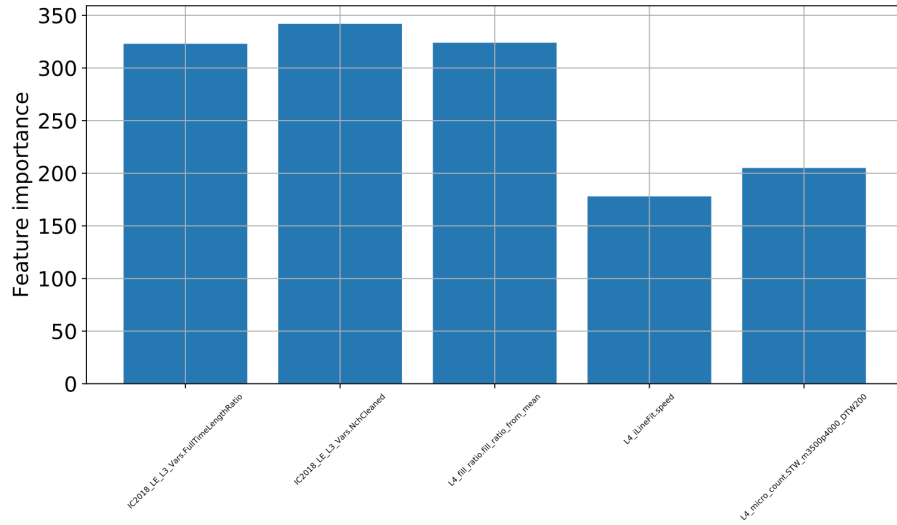


Figure 14: Feature importance of each input variable in the L4 noise classifier.

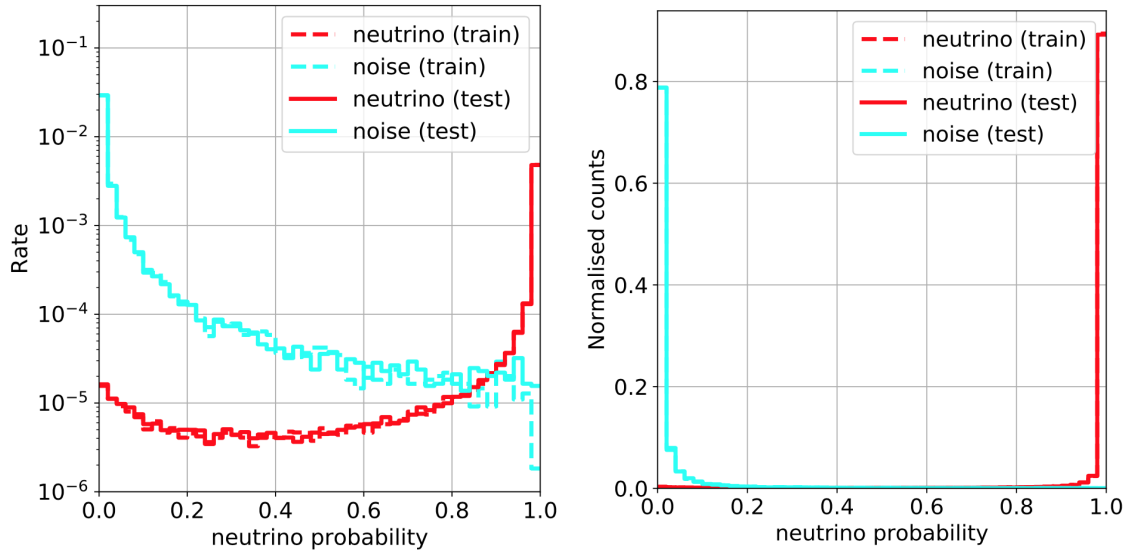


Figure 15: Distributions of L4 noise classifier predictions (probability event is a neutrino) for noise and neutrino events. The left panel show the rates [Hz], whilst the right panel shows the normalised distributions.

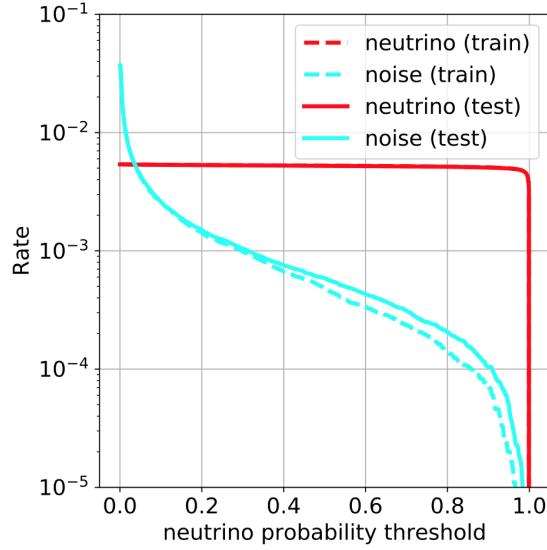


Figure 16: Noise and neutrino rate [Hz] as a function of classifier probability cut value.

3.6.3 Classifier for muon background

The second part of the Level 4 is designed to reject events caused by atmospheric muons, while keeping most of the neutrinos. For this classifier, detector data (following L4 noise cut) is used as the background sample used for training, as it is 99% muons at this stage and is considered more robust than using MuonGun MC at this processing level. The signal training sample is still GENIE MC. This also allows unsimulated event populations (such as muon bundles) to be removed by the classifier. Note that a classifier was also trained using MuonGun MC for the background sample, which achieved similar performance but was not used in the sample.

The background sample is comprised of the following data runs, which were selected to cover years roughly equally in order to avoid strong dependence of the muon rejection on the specific season due to muon flux seasonal variations.

- 2012: 120200, 120700, 121650,
- 2013: 122650, 123250, 124650,
- 2014: 125150, 125700, 126300,
- 2015: 126850, 127400, 127850,
- 2016: 128000, 128550, 129050,
- 2017: 129650, 130150, 130700.

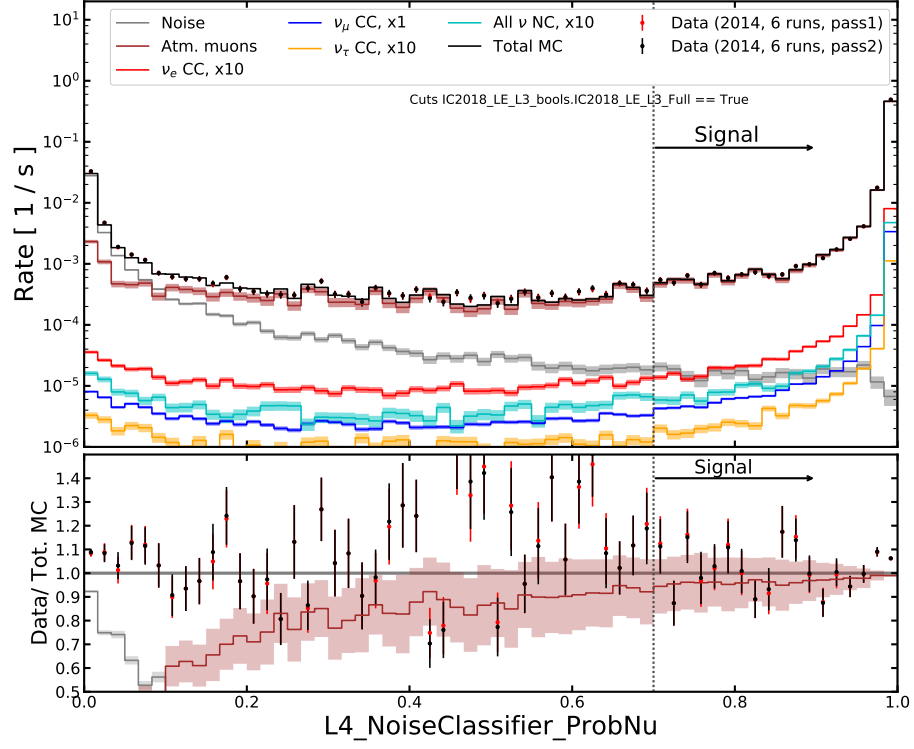


Figure 17: Distribution of L4 noise classifier output for data and MC. The x axis value is the predicted probability of being a neutrino events (as opposed to pure noise).

Table 12: Level 4 muon classifier input variables.

Variable	Description
IC2018_LE_L3_Vars.ICVetoHits	See description in section 3.5.1
IC2018_LE_L3_Vars.RTVeto250Hits	See description in section 3.5.1
IC2018_LE_L3_Vars.NchCleaned	See description in section 3.5.1
IC2018_LE_L3_Vars.NAbove200Hits	See description in section 3.5.1
L4_VICH_nch	Number of triggered DOMs in the Veto Region that can be caused by muons
L4_accumulated_time	Time to reach 75% of an event's charge in the cleaned pulse series. This is one of the few charge-dependent variables in the selection.
L4_first_hlc_rho	Radial distance from string 36 (roughly the center of DeepCore) of the first HLC hit
SRTTWOofflinePulsesDCHitStatistics.cog.z	Depth of the center-of-gravity of all hits in the cleaned pulse series
SRTTWOofflinePulsesDCHitStatistics.z.sigma	RMS of the vertical positions of all hits in the cleaned pulse series.
SRTTWOofflinePulsesDCHitStatistics.z.travel	The total vertical span of the hits in the cleaned pulse series.

Table 12 lists the 10 input variables used in the noise classifier, with the variable distributions for data and MC shown in Figures 18, 19 and 20. The ‘feature importance’ of all classifier input variables is shown in Figure 21, showing that all variable are contributing significantly.

Figure 22 shows that the muon classifier achieves good separation of signal and background, as was the case for the noise classifier. There is excellent agreement between the training and test sets, and therefore no sign of overtraining. The muon and neutrino rates as a function of cut value on the classifier prediction are shown in Figure 23, showing that the muon rate can be reduced by 1-2 orders of magnitude before the neutrino rate drops too steeply.

The classifier prediction is written to the frames as `L4_MuonClassifier_Data_ProbNu`, and its distributions for data and MC are shown in Figure 24. At cut is applied at 0.65, removing 94% of all muon events, while keeping 87% of all neutrinos.

3.6.4 Cut performance

The only cuts made at level 4 are the previously described cuts on the noise and muon classifier predictions, which result in the rejection of over 99% of the pure noise events and

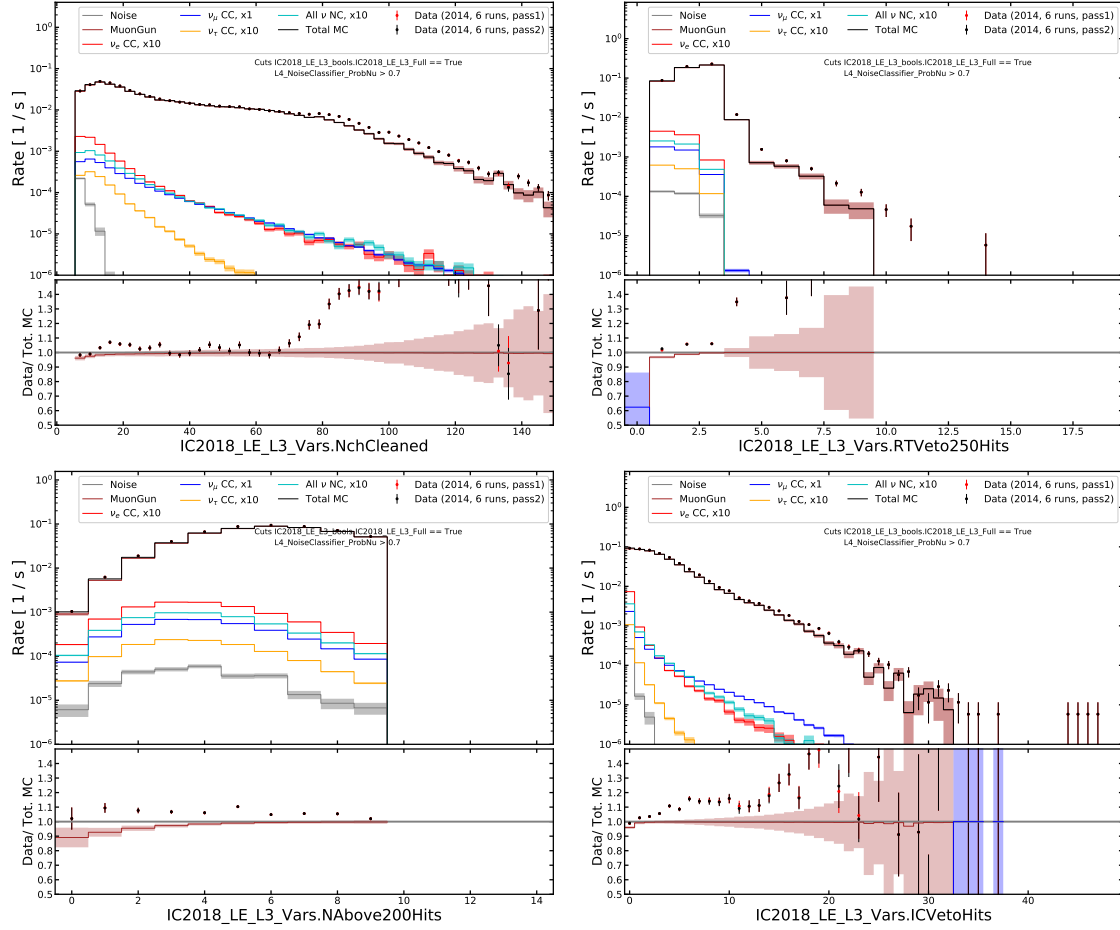


Figure 18: Level 3 variables used as inputs for Level 4 muon rejection BDT: number of DOMs in cleaned pulse series (top left), number of DOMs found by RTVeto (top right), number of hit DOMs above $Z=-200$ m before the trigger time (bottom left), number of hits in veto region (bottom right).

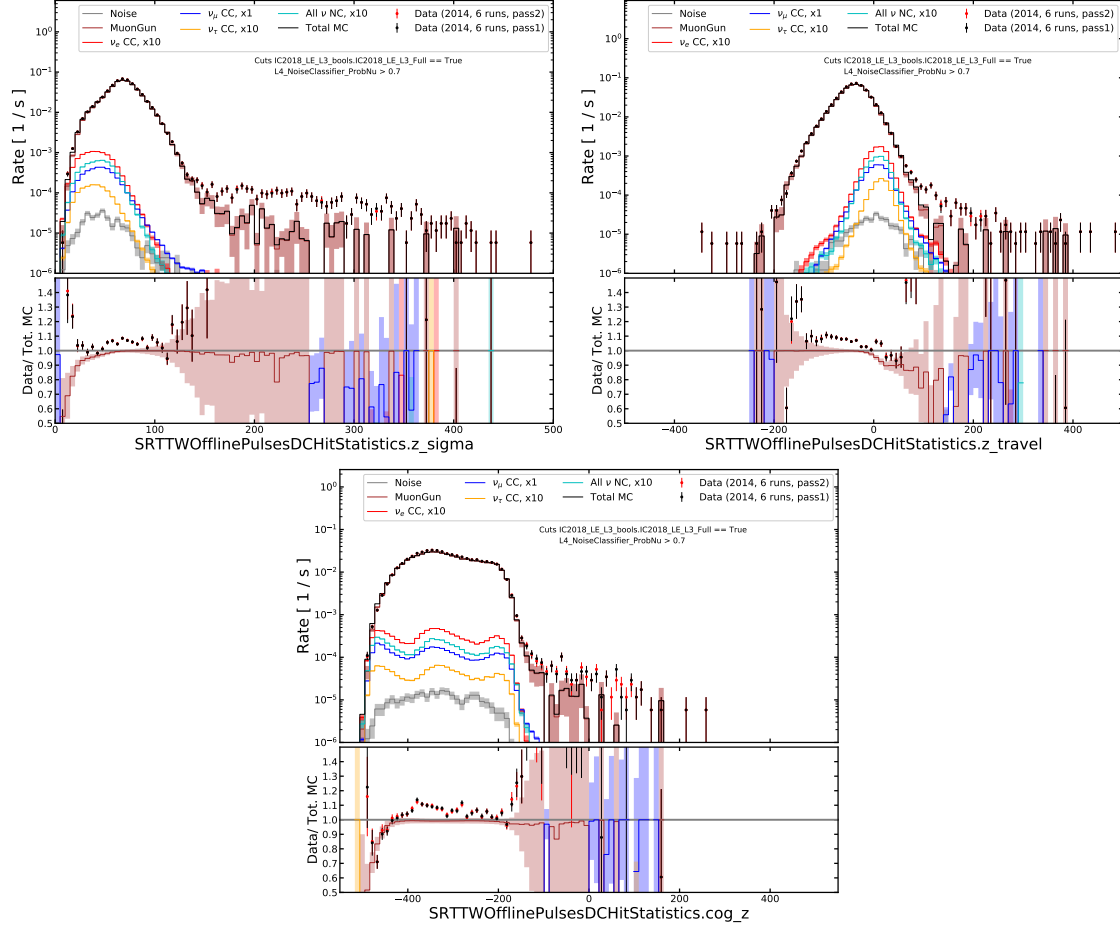


Figure 19: Hit statistics variables used for the Level4 muon BDT classification: Z_{σ} (top left), Z_{travel} (top right) and Z position of the center-of-gravity (bottom).

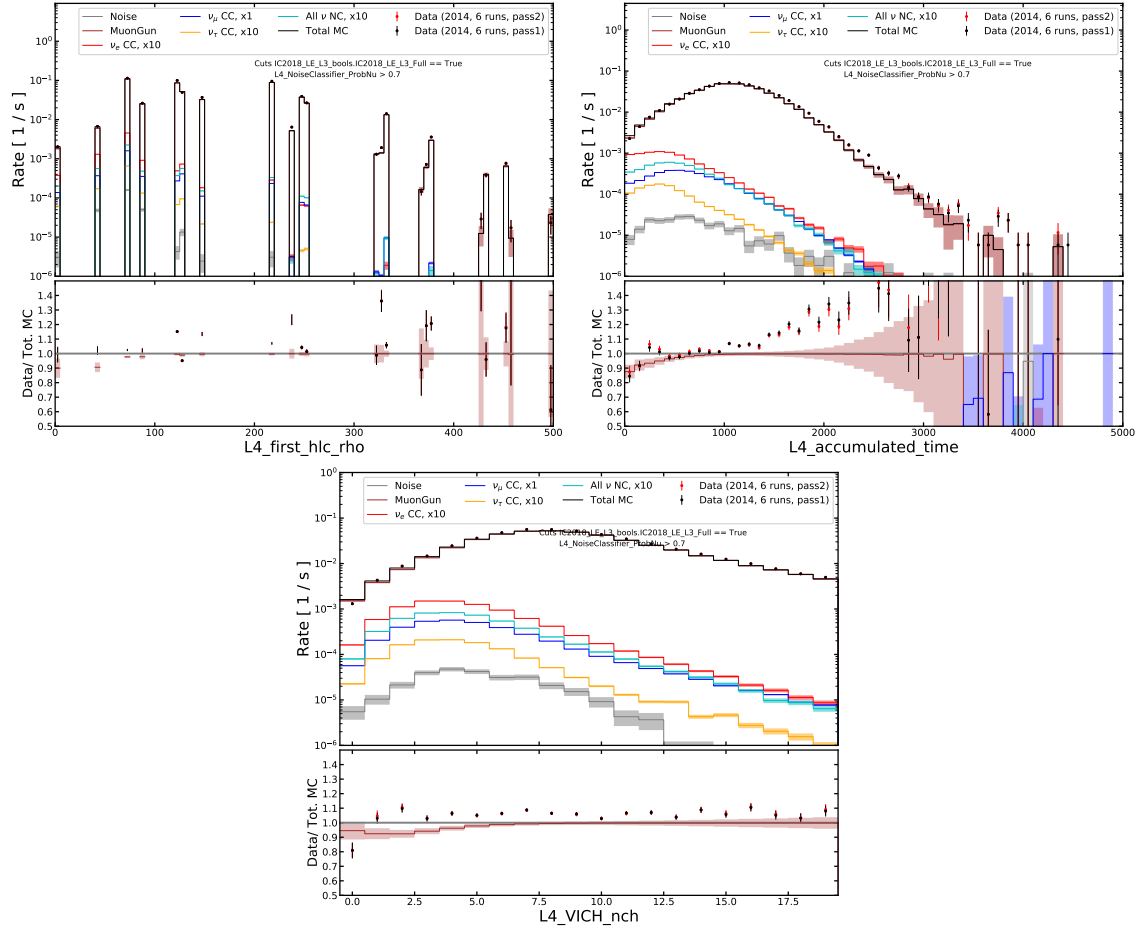


Figure 20: Variables used for Level 4 muon BDT classification: distance from the first HLC hit to string 36 (top left), time to reach 75% of the charge (top right) and number of hits in casual veto region (bottom).

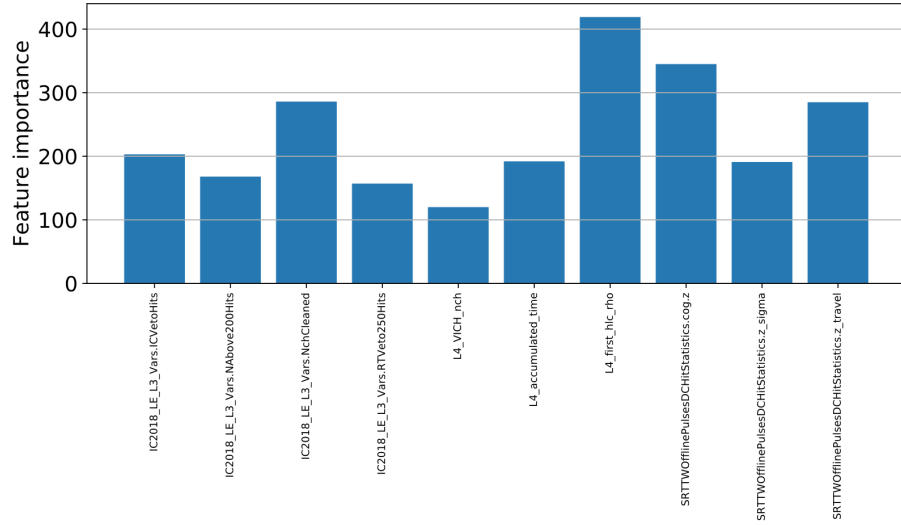


Figure 21: Feature importance of each input variable in the L4 muon classifier.

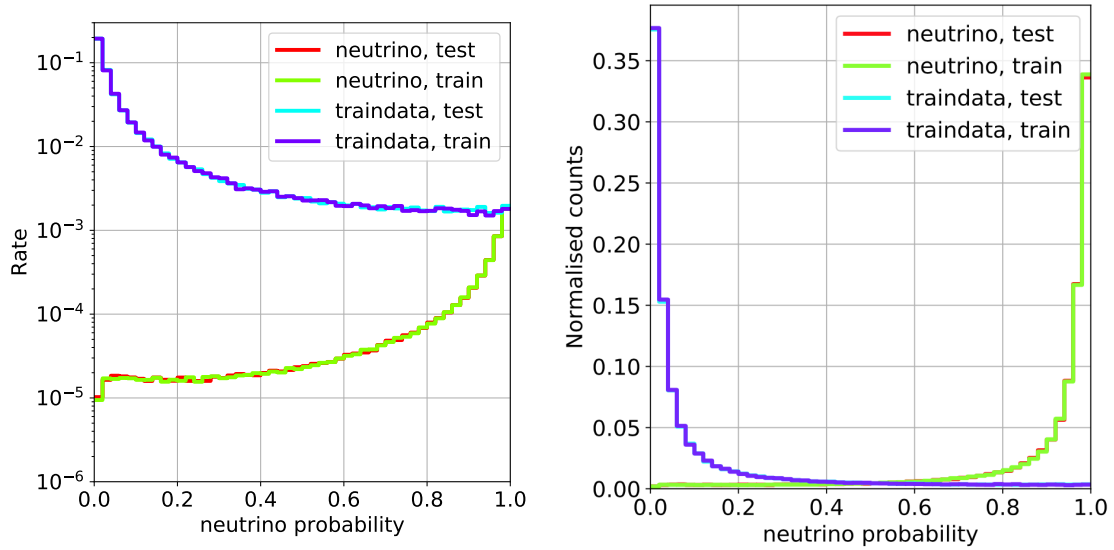


Figure 22: Distributions of L4 muon classifier predictions (probability event is a neutrino) for muon and neutrino events. The left panel show the rates [Hz], whilst the right panel shows the normalised distributions.

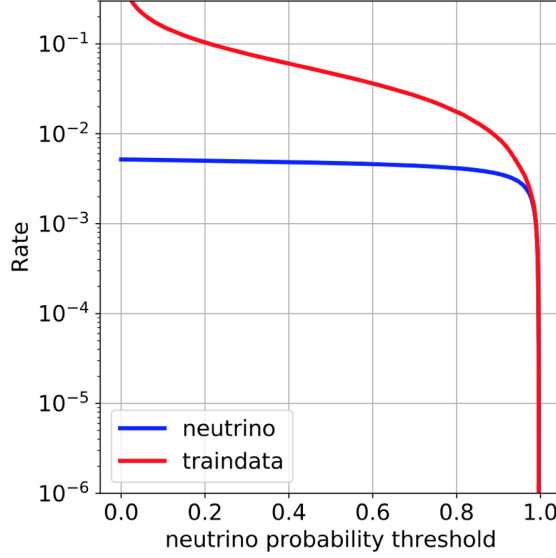


Figure 23: muon and neutrino rate [Hz] as a function of classifier probability cut value.

665 over 94% of atmospheric muon events (relative to the rates before making the L4 cuts).
 666 At the same time, over 80% of all neutrino events are kept. The highest fraction of kept
 667 neutrinos is observed for CC interactions of tau neutrinos with the efficiency over 92%,
 668 whilst the lowest efficiency of 82.5% is seen for CC interactions of muon neutrinos, what
 669 can be described by the similarities between muons produced in neutrino interactions and
 670 from atmospheric background. The resulting sample after level 4 cuts has roughly 100:10:1
 671 muon:neutrino:noise rates, with the MC rates being listed in Table 13.

672 The comparison of the event rates before and after Level 4 selection, as well as the
 673 selection efficiency, are shown in Figure 25 as a function of neutrino energy and (cosine)
 674 zenith angle.

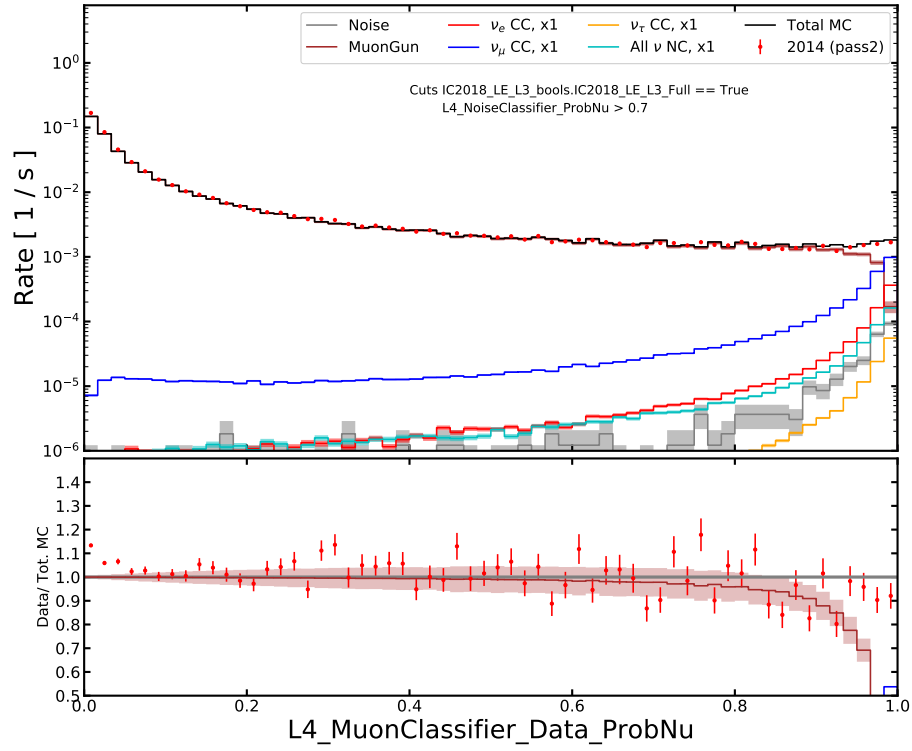


Figure 24: Distribution of L4 muon classifier output for data and MC. The x axis value is the predicted probability of being a neutrino event (as opposed to a muon event).

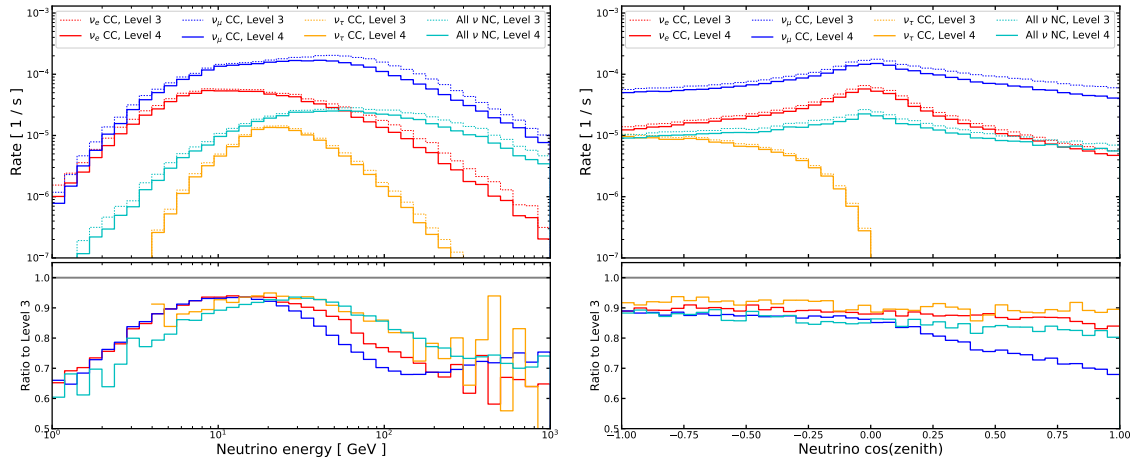


Figure 25

Table 13: Event rates at DeepCore Level 3 and after Level 4 noise and muon rejection BDTs, and corresponding efficiency of the level 4 cuts (relative to level 3).

	Level3 [mHz]	Level 4 (noise) [mHz]	Level 4 (full) [mHz]	Level 4 efficiency [%]
GENIE, ν_e CC	0.95	0.90	0.84	88.5 %
GENIE, ν_μ CC	3.77	3.65	3.11	82.5 %
GENIE, ν_τ CC	0.129	0.124	0.119	92.1 %
GENIE, ν NC (all)	0.53	0.51	0.46	85.5 %
MuonGun, atm. μ	505	490	28.1	5.6 %
Vuvuzela, pure noise	36.6	0.28	0.28	0.7 %
Total MC	547	495	32.9	6.0 %
Data (2014, pass2)	582.0	526.0	31.6	5.4 %
Data (2014, pass1)	581.6	525.7	31.5	5.4 %

3.7 Level 5

The main goal of Level 5 selection [script] is to further reduce the remaining muon background, achieving a neutrino dominated sample. At this stage of processing, the remaining muons are those that do not leave clear muon-like signatures, and as such have evaded detection thus far.

The Level 5 selection cuts can be split into two categories: **Starting Containment** cuts have the objective of removing events occurring outside or at the edge of the DeepCore fiducial volume, whereas **Corridor** cuts are targeted to remove *sneaky muons*, which can pass through the veto regions undetected along specific corridors in the hexagonal IceCube grid structure without hitting any IceCube strings.

The **corridor cut module** works by first defining a set of poorly-instrumented corridor directions for every DeepCore string along which a muon can travel. Given an uncleaned pulse series, the COG of the event is calculated, along with the closest DeepCore string associated with it. Given that string, the algorithm looks for any triggered DOMs (hits) located along any of the identified corridors (basically azimuth directions), within a cylinder of radius R and a time window of size $[w_{min}, w_{max}]$. For the Level 5 of oscNext, the value of these parameters have been set to be more loose than in previous event selections as the performance was found to improve, Table 14 lists the default values used. Ultimately, cuts are made based on the number of DOMs with detected light near to the identified corridor.

An infinite track reconstruction **SPEFit11** is also run at level 5. This is used to seed final level reconstructions at the next processing level. Another use is that the reconstructed track direction can be compared to the corridor direction, and if they disagree significantly

Table 14: L5 Corridor Cuts parameters

Input	Value
Radius	250 m
WindowMinus	-1000 ns
WindowPlus	1000 ns
ZenithSteps	0.02 rad

Table 15: Level 5 cuts.

Overview	Description
Radial containment	Keep events within 150 m of the center of DeepCore (string 36), considering three estimates of the vertex (brightest string, first HLC hit position, level 3 vertex guess)
Vertical containment	Keep events within $z=[-490,-220]$ m, considering two estimates of the vertex (first HLC hit position, level 3 vertex guess)
Corridor cut	Remove events that have > 2 hit DOMs along the brightest corridor (pre-defined) or have a cosine of the angular separation between the corridor direction and SPEFit11 direction of > 0.7 rad.
Tighten classifiers	Apply stricter cuts to the level 4 noise (>0.85) and muon (>0.9) classifiers

697 this suggests that the corridor light observed may instead be due to noise coincident with
698 a neutrino, rather than a sneaky muon. This check is used to minimize the accidental
699 removal of signal events.

700 The cuts made using these variables are described in Table 15, and the data and MC
701 distributions for the variables shown in Figures 26, 27, 28 and 29.

702 3.7.1 Event rates and performance

703 The event rates at level 5 are shown in Table 16. A neutrino dominated sample is achieved,
704 with an approximate neutrino:muon ratio of 2:1. The neutrino selection efficiency vs energy
705 and (cosine) zenith angle is shown in Figure 30. High energy events are lost here due to
706 the containment cuts.

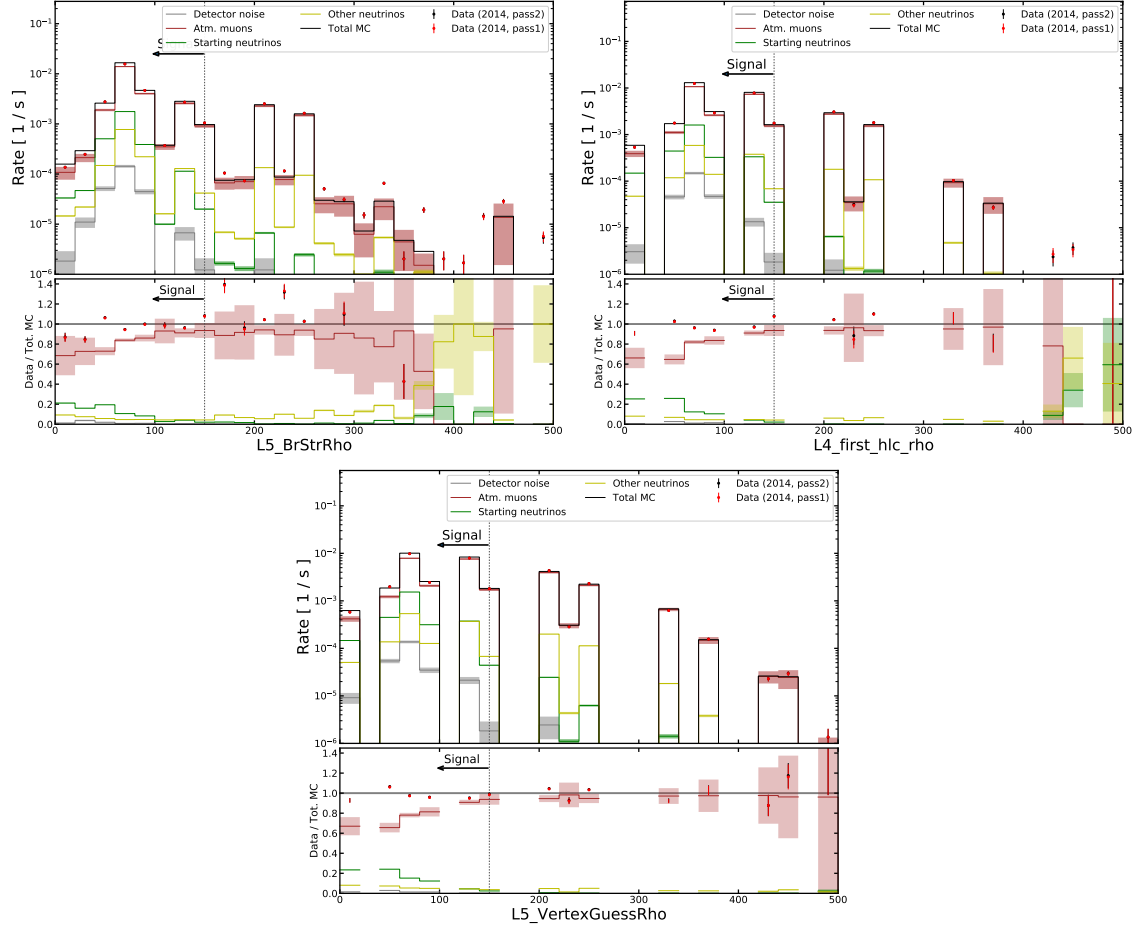


Figure 26: Distribution of variables used for radial starting condition: the brightest string ρ (top left), the first HLC ρ (top right) and ρ of the first pulse in the cleaned pulse series (bottom).

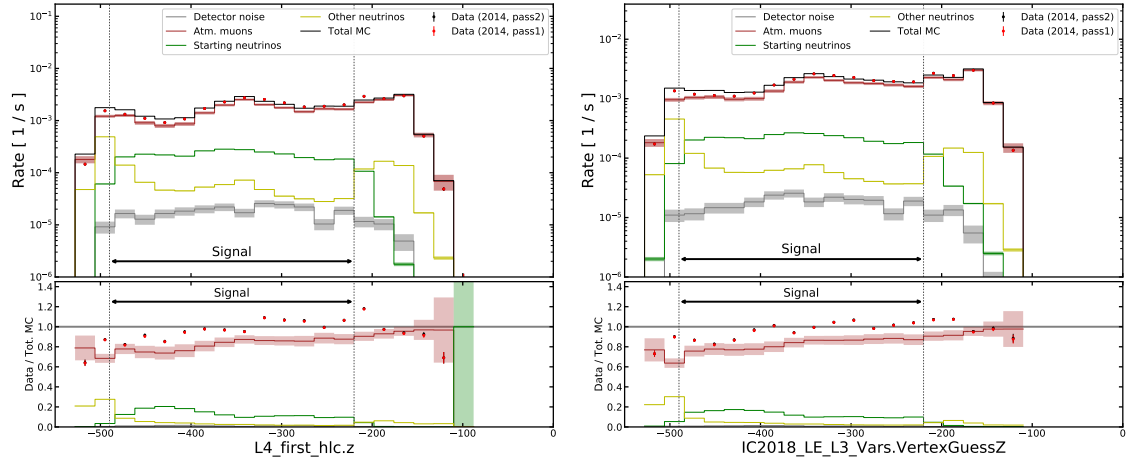


Figure 27: Distribution of variables used for depth starting condition: Z-position of the first HLC pulse (left) and Z-position of the first DOM in the cleaned pulse series (right).

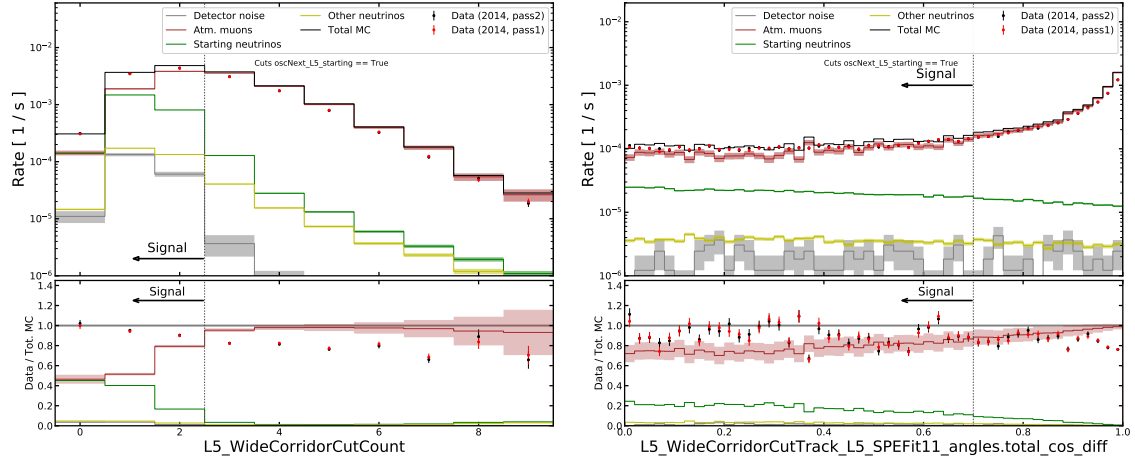


Figure 28: Number of DOMs found in L5 corridors (left); cosine of an angle between the brightest corridor and SPEFit11 direction (right)

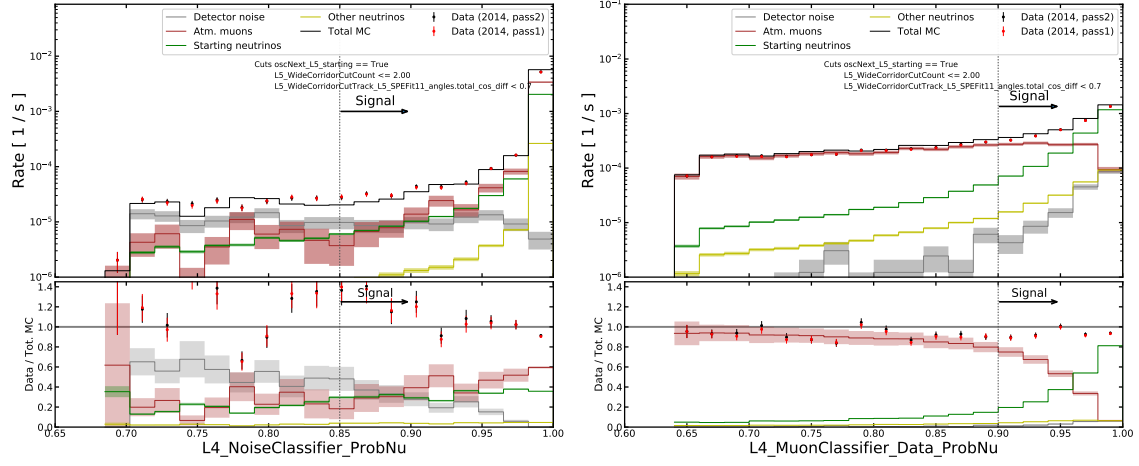


Figure 29: Distribution of Level 4 noise (left) and muon (right) BDT scores after starting and corridor cut requirements of Level 5.

Table 16: Passing Rates at OscNext Level 5

	Level 4 [mHz]	L5 (contain) [mHz]	L5 (corridor) [mHz]	L5 (full) [mHz]	L5 (efficiency) [%]
Signal					
ν starting	2.89	2.60	2.22	1.95	67
ν other	1.63	0.39	0.29	0.21	13
Components					
ν_e , CC	0.84	0.61	0.53	0.48	57
ν_μ , CC	3.11	2.00	1.64	1.39	45
ν_τ , CC	0.12	0.086	0.078	0.071	60
ν , NC	0.46	0.31	0.26	0.23	49
Background					
Atm. μ	28.1	11.8	3.09	0.93	3.3
Noise	0.26	0.21	0.18	0.07	27
All MC	32.89	14.99	5.77	3.16	9.6
2014, pass1	32.29	14.29	5.82	3.17	9.8
2014, pass2	32.43	14.34	8.88	3.15	9.7

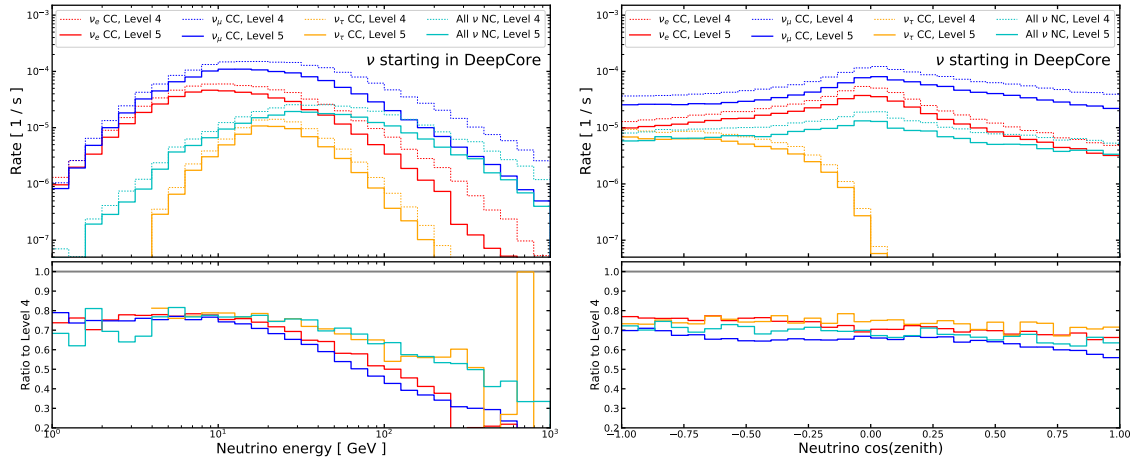


Figure 30: Selection efficiency for Level 5 selection as function of neutrino energy (left) and zenith angle (right).

4 Reconstruction and PID

After level 5, background and signal rates have now more comparable rates, and the total number of events left in the selection can be reconstructed with more complex algorithms with a reasonable amount of computing resources. At this point, diverging paths are taken by the two samples: the *verification sample* uses **SANTA/LEERA** to reconstruct events (for details see the [oscnex verification sample note](#)), while the *high statistics sample* uses **RetroReco**. [Link to high stats sample note](#).

The oscNext sample is almost entirely dominated by neutrinos below 1 TeV, and selects events starting in the DeepCore volume. A suitable reconstruction hypothesis is therefore a cascade + finite track (where the track may be vanishing for events with no final state muons). Stochastic energy losses from muons are low in this energy range, and muons may be considered to be Minimum Ionising Particles (MIPs).

A number of algorithms for reconstructing such a hypothesis have been developed over the years:

Pegleg was the low energy event reconstruction used for the previous generation of oscillation analyses. A table-based eight-dimensional likelihood function is minimized using the MultiNest seeded nested sampling algorithm for six of the eight dimensions². The underlying hypothesis is a charged current muon neutrino interaction, resulting in the general hypothesis of a cascade and outgoing track. Track- and cascade energy are optimized internally for each evaluated six-dimensional hypothesis. The interaction type (mostly: cascade or track) can be determined after reconstruction, based e.g. on the reconstructed track vs cascade energy fraction. Pegleg can be expected to be able to reconstruct all events within a standard low-energy sample with reconstruction times at the order of minutes per event.

Retro is a similarly efficient reconstruction as Pegleg and will be used as the default reconstruction for OscNext, due to the better resolutions and/or run-times it yields. The approach is also a table-based likelihood minimization, that is, other than Pegleg, light-source-based. It performs a seed and main optimization and gives information on all parameters of the reconstructed interaction.

SANTA is a relatively simple and therefore very fast but less efficient way to reconstruct direction and vertex of an interaction by fitting a Cherenkov cone to directly hit DOMs, direct hits being ones from unscattered photons.

LEERA is a table-based energy reconstruction algorithm. It uses SANTA and finitereco output to seed the event direction and vertex. The algorithm then scales the energy in the cascade and the track until the hit-no-hit probability is maximized across all DOMs. The reconstruction time per event is at the order of seconds.

²These six dimensions are time, three spacial dimensions and two es.

743 A comprehensive study comparing the different methods was performed (see section G),
744 resulting in the selection of SANTA/LEERA as a simple and fast reconstruction for the
745 verification sample (see the [oscnext verification sample note](#)), and RetroReco as the highest
746 performance reconstruction that was used for the high stats sample.

747 PID is also crucial to oscillation analyses, as identifying track vs cascade events is
748 sensitive to the dominant $\nu_\mu \rightarrow \nu_\tau$ oscillation channel in $\mathcal{O}(10 \text{ GeV})$ atmospheric neutrino-
749 nos. Machine learning based classifiers are developed using the final level reconstructed
750 properties for both the verification and high stats samples.

751 Appendix A Reviewer Q and A

752 Reviewer comments/questions pertaining to the simulations and common sample (to and
753 including level 5) are listed below, and are continually being updated and responded to.
754 Questions related to the higher level samples and physics analyses are listed in their re-
755 spective notes.

756 A.1 Ben Jones (WG reviewer)

757 Ben's comments and analyser responses can be found in this [document](#).

758 A.2 Carlos de los Heros (Collaboration reviewer)

759 Have so far read sections 1 and 2 of version 03 of the note.

760 1) Just for my education (no changes needed in the document), in line 137 it is men-
761 tioned that the neutrino sample is simulated with 10x the livetime of the data. Is that a
762 general aim to be kept for future analyses, and how the continuously increasing livetime of
763 the data will affect the production of this background?

764 [Tom] We do expect to maintain 10x livetime in the future. If this becomes prohibitive
765 we will need to re-think, either by quantitatively evaluating the impact of lower MC livetime
766 on the analyses, or by developing more efficient MC production methods (for example the
767 KDE prescale method used with MuonGun).

768 2) It would be useful to specify what definition of DeepCore you use, and add the typical
769 figure of the surface placement of the strings marking which area you consider DeepCore.
770 I would suggest to add a scale so the reader has a sense of the size of the generation and
771 tracking volumes you talk about later in tables 1 and 2.

772 [Tom] I have added a section in the appendix showing the fiducial volume.

773 Section 1.3

774 3) It should be stated explicitly here for an external reader whether you aim for an
775 all-sky analysis (with corresponding all azimuth simulations) or not. You select starting
776 events, which point to an all-sky analysis, but later in line 164 you mention up-going events
777 in the verification sample. This can be confusing.

778 [Tom] I have now clarified this in the note. The sample overall features events from
779 all direction, so is all-sky. Individual analysers are free however to cut out the down-going
780 reason in their own final cuts, which has been done by the verification sample team. This
781 is to (a) reduce muons (which is instead achieved by a classifier in the high stats sample)
782 and (b) to avoid dealing with self-veto effects, which have not been quantitatively studied
783 at these energies.

784 4) Event selection stages (specifically lines 158/159): why isn't L5 applied before L4?
785 I would assume that removing corridor muons will be just some straight cut, which would
786 reduce the amount of data to treat with the BDT later.

787 [Tom] The corridor cut module is computationally intensive (ms per event), and would
788 be very slow to run over all events at L4, so we wait for L5 where the rates have dropped
789 by a further order of magnitude. This module is not written for performance though, and I
790 expect we will speed this up significantly and probably move it lower in the chain in future
791 analyses (probably when we update this module for Upgrade compatibility).

792 Section 1.5

793 5) neutrino flux and muon flux: aren't these correlated and the systematic should be
794 dependent on the systematic of each other?

795 [Tom] They are. We would like to add a correlated treatment of the neutrino and
796 muon flux systematics, but have not yet had time. This would require using MCEq to
797 also generate the muon surface flux, and then modelling the energy losses of the muons
798 through the Earth. Since the muon rate is so low that it's flux systematics are essentially
799 negligible to the analyses, we have not done this yet and proceed with the (conservative)
800 approach of treating them as independent (re-using the muon flux treatment from the
801 previous generation of analyses). As a side note, we plan to add a muon sideband to our
802 DeepCore analyses in the near future to fit simultaneously with neutrino data to constrain
803 both the air shower and detector systematics, so this work will be done as part of that
804 project.

805 Section 2.

806 6) I noticed that the scripts linked have the year 2009 hardwired. Does this have any
807 effect at all?

808 [Tom] This is for the date in the event header for MC events, which is not used for
809 anything and thus arbitrary.

810 7) line 262: strictly speaking the thermal noise comes from the PMT itself (mainly
811 the photocathode, some from other material inside like the electrodes), not from photons
812 hitting the PMT (which suggests external photons)

813 [Juan Pablo] That is correct. The current text is wrong, and has been changed to:
814 comes from spontaneous emission of single electrons from the photocathode due to its
815 thermal energy. The emission is uncorrelated among PMTs.

816 8) Figure 4: You mention correctly that there are two sources of non-thermal noise,
817 and one is radioactive decays in the ice. But the detection of light from these decays will
818 be affected by the DOM acceptance, so this kind of noise should enter the simulation one
819 step before. I know that this might be impossible to implement (do we even know what
820 fraction of non-thermal noise is due to ice activity? I guess not)

821 [Alex] The dominating source of non-thermal noise are decays in the pressure vessel
822 glass, rather than the ice. I corrected the text to reflect that. On the question of calibration:
823 the noise rates and shape of time distribution are calibrated using in situ data, so they have
824 the "correct" DOM efficiency baked in. Systematic uncertainties still exist in the shapes
825 of the SPE templates and how they interact with changes in the effective DOM efficiency,
826 which are part of the detector simulation.

827 9) line 313: it seems that something would come after this line, an enumeration of the

828 two sources of noise. However the sentence ends here abruptly.

829 [Alex] That is correct, the text that the line was originally referring to was moved to
830 section 2.2(.0). The line is removed now.

831 10) line 323: Energy simulated from 0 TeV? (line 337 says from 1 GeV to 10 TeV)

832 [Joshua] Yes, this was a typo and was meant to be 10 TeV. It is fixed in the new version.

833 11) Table 1 and 2. How does someone make sure that the chosen volumes are optimal
834 and we are not losing events that otherwise would trigger? How the optimization was
835 done? Do there exist some plots that show triggers as a function of volume size or something
836 on those lines?

837 [Kayla] A smaller test sample was produced that included a much larger volume than
838 was necessary. By looking at the distribution of events that survive to L5 (slide 25 here),
839 we can see what parts of the volume are necessary. The edges of the volume are chosen
840 such that few to no events are lost by L5. A larger simulation set is then produced with
841 the chosen settings.

842 Section 3

843 12) I went through chapter 3, which is a bit difficult to evaluate for someone who did not
844 participate in the analysis. I do not have any real comment. I know that one can not expect
845 complete MC/data agreement at this point, but I wonder what the, quite strong sometimes,
846 discrepancy between data/MC in practically all the figures in this section mean. Should I
847 worry about it? I guess not because one reaches agreement in further analysis levels. But
848 after looking at all the figures one remains with the question of whether the discrepancy
849 is telling something already at this point (besides lack of bundles in the simulation), and
850 if there could be something to do to bring data/MC agreement closer already at this level.
851 It is rather a philosophical question, I am not requiring any changes or tests.

852 [Tom] See answer to Ben's question 15

853 13) Figure 17. How was the cut point chosen?

854 (Tom) This cut value reduces the noise rate to 10x below the neutrino rate (see Fig
855 23), without much of a loss of neutrino rate. The cut can be pushed harder before the
856 neutrino rate starts to drop quickly, and harder cuts of this classifier are made at final level.
857 At L4 we chose to be conservative and not cut too hard to avoid making the sample too
858 dependent on this classifier (which is a new development, we have not trained classifiers on
859 noise MC in the past), but at final levels we have verified we can achieve good data-MC
860 agreement with harder cuts and so make them.

861 14) line 683 and tables 12 and 15. It is not clear with respect to what the efficiency is
862 quoted, with respect to the previous level, or globally? Specify.

863 [Tom] I have specified this in the text now. In all cases, the efficiency of a processing
864 level is shown relative to the preceding level (e.g. L4 efficiency of % of L3 events kept by
865 L4 cuts)

866 Typos and such:

- 867 • line 269: vuvzela $\rightarrow$ vuvuzela - fixed

- 868 • caption fig 2: shouldn't it be that the scaling is only applied to the orange (not
869 yellow)? - the yellow curve is the red curve + scaling, now clarified in the note
870 caption fig 3: constantsl → constants - fixed
- 871 • caption fig 6: live-tme → live-time - fixed
- 872 • line 343: radios → radius - fixed
- 873 • line 399: operations → operates - fixed
- 874 • line 406: based ON many factors, (note the comma) - fixed

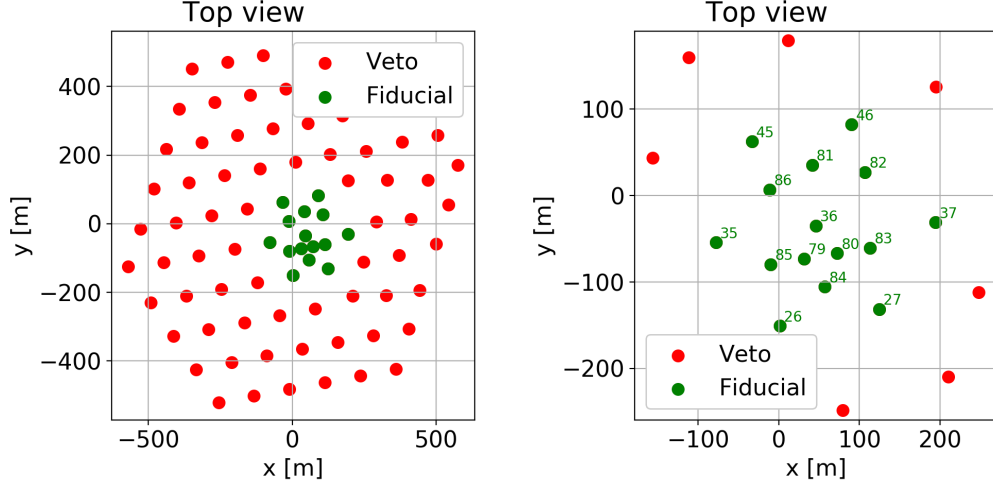


Figure 31: Strings within the DeepCore fiducial volume, as viewed from above. The left panel shows the full detector, whilst the right panel shows a close-up of the DeepCore region, where the fiducial string numbers are indicated.

Appendix B Fiducial volume

Figures 31 and 32 indicate the strings/DOMs that form the DeepCore fiducial volume used in this analysis, which is defined by $-490 < z < -220$ and $\rho_{36} < 150$ (a 20.5 Mton volume). This is the containment volume applied to the vertex estimates in L5 (see section 3.7). Note that vertices are required to fall within this volume, but light on DOMs outside this region is allowed. Note also that some algorithms used throughout their event selection may use differing definitions of what precisely constitutes the veto region.

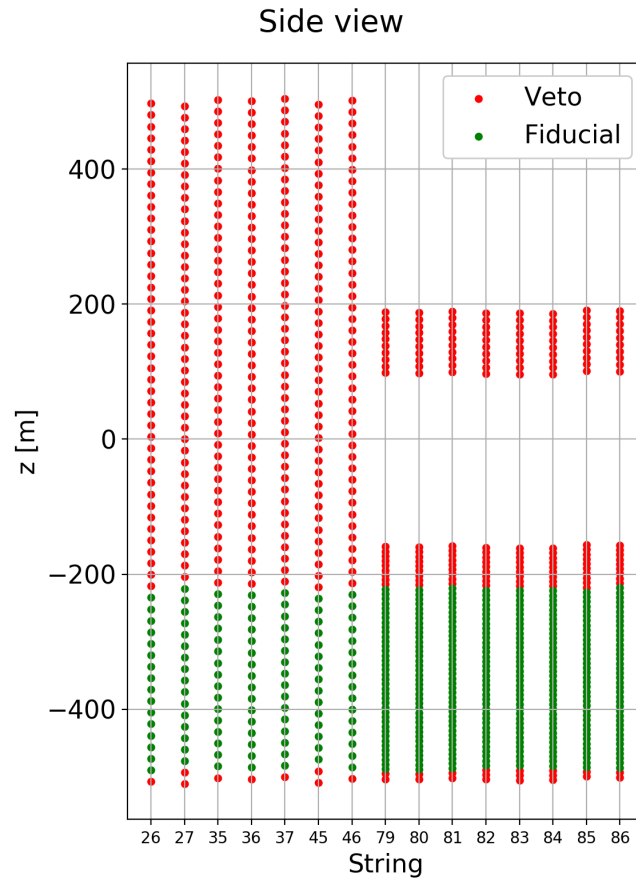


Figure 32: DOMs within the fiducial volume on DeepCore fiducial strings, as viewed from the side.

Table 17: oscNext metaproject releases.

Release	Description
v01_00_01	Version used prior to update of DOMLauncher to fix SLC sim bug.
v01_00_02	Same as v01_00_01 but with latest DOM-Launcher trunk to fix SLC sim bug.

882 Appendix C Software

883 A dedicated `oscNext metaproject` containing all IceTray software projects (and versions)
884 required to run oscNext simulation, event selection and reconstruction has been created.
885 **All oscNext users should use this.** Releases of the metaproject are listed in Table 17.
886 Installation instructions are provided [here](#). Users **must** use the `py2-v3.1.1` environment
887 on CVMFS with this metaproject.

888
889 Pre-built versions of the metaproject are deployed to CVMFS, with the latest being
890 `/cvmfs/icecube.opensciencegrid.org/users/Oscillation/software/oscNext_meta/releases/latest/`
891

892 The majority of the code developed specifically for the oscNext event selection is in the
893 `oscNext icetray project`. See the enclosed [README](#) for usage instructions.

894
895 The high level event sample processing/submission and verification scripts, plus machine
896 learning training/testing scripts, live in the `fridge repository`. Instructions are [here](#).

Table 18: Simulation frame objects

Object	Description
<code>frame["I3MCWeightDict"]["weight"]</code>	Nominal weight for this event, normalised for that file. Gives an event rate in [Hz]. Must divide by number of files used to get correct weight. Uses a nominal flux and oscillation model (Honda 2014 + NuFit v2 for neutrinos).
<code>frame["I3MCWeightDict"]["MCInIcePrimary"]</code>	Particle containing the in-ice primary particle (e.g. neutrino for GENIE and muon for MuonGun). Not present for noise MC.
<code>frame["I3MCWeightDict"]["MCAirShowerPrimary"]</code>	If available, also have information on the primary cosmic ray producing the air shower (mainly in CORSIKA)
<code>frame["MCDeepCoreStartingEvent"]</code>	Boolean flag indicating if the event interaction vertex is within the DeepCore fiducial volume (GENIE only)
<code>frame["MCExtraTruthInfo"]</code>	A map containing extra useful truth information, like vertex ρ_{36} , $\nu_{\mu,CC}$ muon track length, true track and cascade energy, etc.
<code>frame["I3GENIEResultDict"]</code>	Map containing information about the interaction from the GENIE event generator. Requires <code>icecube.genie_icetray</code> to be imported to load (this is available in the oscNext metaproject).

Appendix D Simulation

D.1 Frame objects

Table 18 lists useful frame objects for working with oscNext simulation.

D.2 MuonGun non-final level datasets

In addition to the MuonGun datasets used in the analysis, a number of sets were also generated and used at earlier stages of the event sample development. These are listed here (unless otherwise specified, parameters used are the same as Table 2:

- 139008

905 Full detector set (e.g. no target cylinder)

906 Wide energy range (50 GeV - 10 TeV)

907 This set gives a good representation of detector data at L3 (muons far from
908 DeepCore still trigger DeepCore due to coincident noise), and is used for training the
909 L4 muon classifier.

910 • **139010**

911 Is similar to the nominal set **139011** but does not use a KDE pre-scale and uses
912 a large target cylinder (radius = 200 m, height = 700 m)

913 This set gives a good representation of detector data at L5, and was used for
914 developing L5 cuts.

915 **D.3 Systematics sets comparisons**

916

Add plots
comparing
systemat-
ics sets at
final level
(Kayla)

917 **Appendix E Detector data**

918 **E.1 Detector seasons**

919 Relevant details on IceCube seasons:

- 920 • 2011

921 First year of DeepCore data

922 Changing online software versions and high initial noise rates mean that the early
923 part of the 2011 season is not used for data analysis

- 924 • 2015

925 Individual DOM charge calibration used for the first time

926 Requires use of pass2 data and individual SPE templates in MC

- 927 • 2017

928 Update to individual DOM SPE charge templates

929 Charge is quantised in reco pulses in detector data from this point onwards

- 930 • 2019

931 Updates in 2019 season vs 2018 (from John Kelley):

932 Updated DOM calibrations and configurations

933 One previously semi-dead DOM (30-60 “Rowan”) returned to data-taking

934 Added new online GRECO realtime low-energy event stream

935 Added new IceAct trigger and filter

936 Updated PnF internals

937 **E.2 Excluding early 2011 runs**

938 Figure 33a and 33b show the level 5 rates of the 2011 and 2012 seasons, including the
939 sudden jump that occurred around run 118500. After inspecting the region around the
940 jump, **run 118550** was selected as the official cutoff for oscNext. Including 2011 data
941 above that run produced rate fluctuations in line with other season-to-season variations,
942 as shown in figure 34.

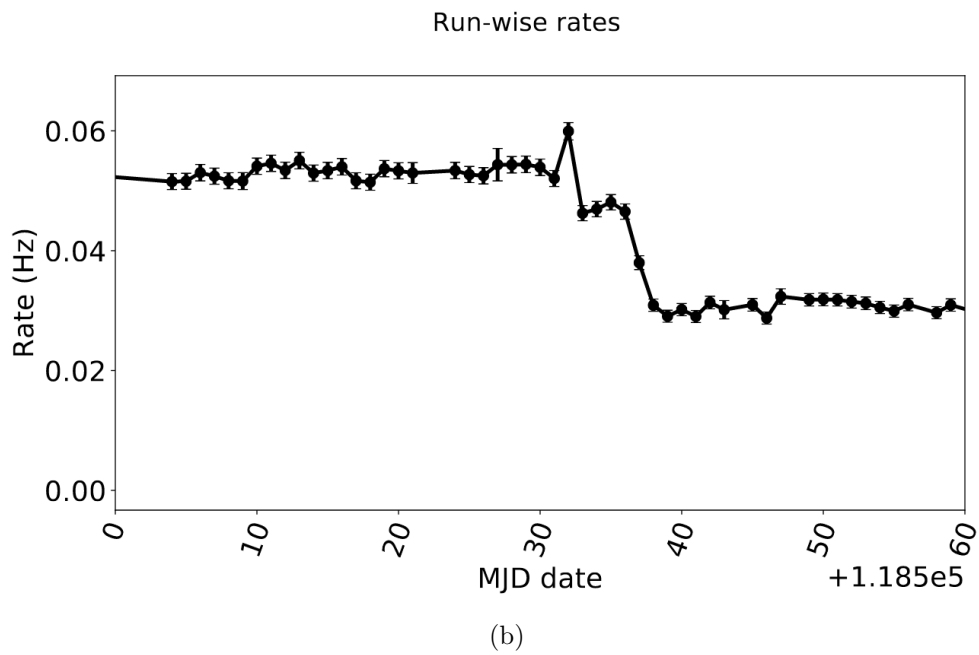
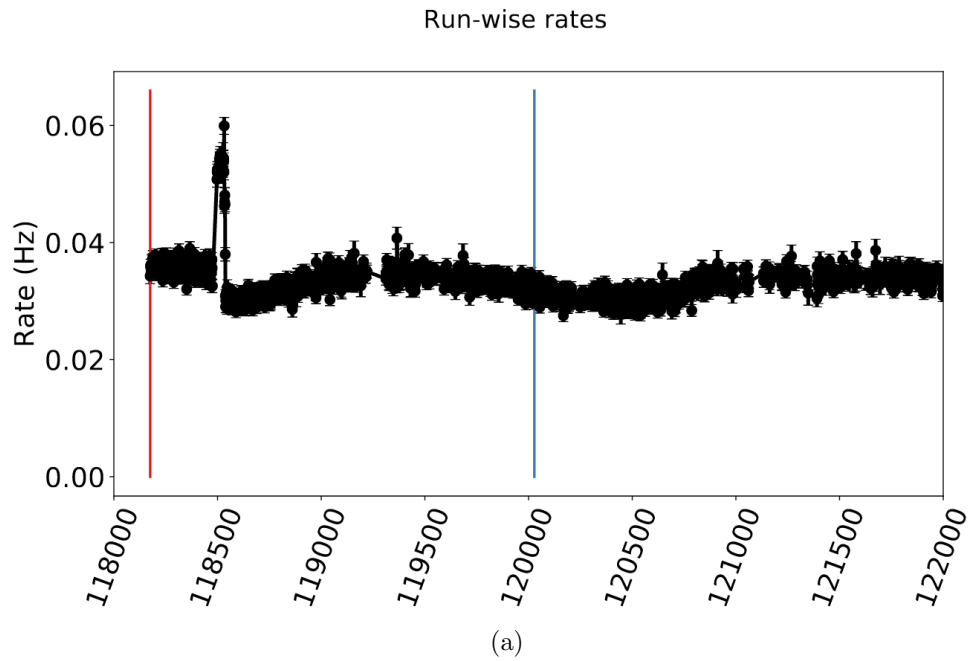
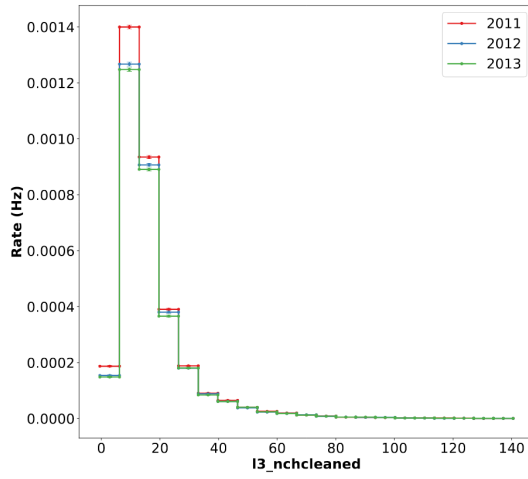
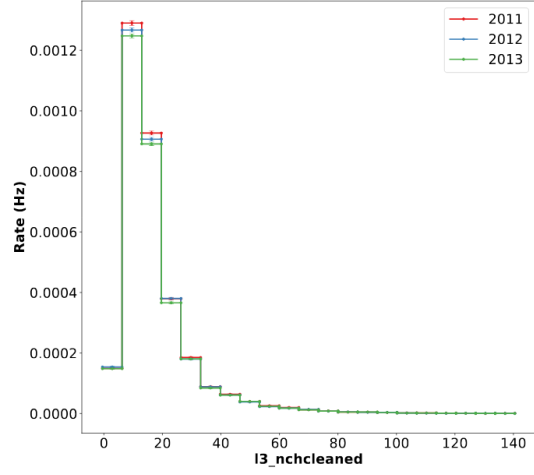


Figure 33: Detector data rates at Level5 in 2011/2012. Vertical lines show the official season starts (red: 2011, blue:2012). Bottom plot shows a zoomed in view of the rates at the 118500 peak.



(a)



(b)

Figure 34: Effect of including the full 2011 sample (left) vs. including only runs above 118550 on a low-level variable (in this case: the number of triggered DOM in the cleaned pulse series). The fluctuations seen in each year is comparable after applying the run start cutoff

943 Appendix F Event sample

944 F.1 Versioning

945 The processing levels are versioned with a major and minor version number. Below are
946 notes on the versioning:

947 • L3

948 **v01.02** - Added data quality cut (SLOP trigger, Light In Detector, e.g. flaring
949 DOMs)

950 **v01.03** - Fixed simulation event headers. Added single power law flux weight
951 (no oscillations) for GENIE.

952 • L4

953 **v01.01** - Noise classifier trained using 888003 and 1?9002 datasets. Muon clas-
954 sifier trained using 1?9002 datasets and detector data. L3 v01.01 as input.

955 **v01.02** - Same as v01.01 but with L3 v01.02 as input.

956 **v01.03** - Same as v01.02 but with L3 v01.03 as input.

957 • L5

958 **v01.02** - Single version of SANTA cleaning selected. Note that these are not
959 used in cuts so these files are functionally identical to v01.01. L4 v01.01 as input.

960 **v01.03** - Same as v01.02 but with L4 v01.02 as input.

961 **v01.04** - Same as v01.03 but with L4 v01.03 as input.

962 • L6 (high stats)

963 **v01.00** - First version. Runs RetroReco and SANTA. Used `/cvmfs/icecube.opensciencegrid.org`
964 `12-17_v01`

965 • L7 (high stats)

966 **v01.01** - Uses variables from L6 v01.00. Uses the prototype classifier V0 as the
967 output variable.

968 **v01.02** - Final reconstructed variable have been renamed according to a common
969 scheme (`L7_reconstructed_variable`). Implemented scaling factor for hadronic cas-
970 cade energy. Classifier output is still based on classifier V0.

971 **v01.03** - Classifier output now comes from version 2 of the BDT.

972 **v01.04** - Added reco energy and track length corrections, PID and photon speed
973 variables. Used `/cvmfs/icecube.opensciencegrid.org/users/stuttard/oscNext.meta/releases/2`
974 `03-27_v01`.

Table 19: Average processing time for GENIE simulation (dataset 140000) for each oscNext processing level. Numbers are approximate.

Level	Events per file	Time per file [min]	Time per event [ms]
L3	13000	5	20
L4	8000	2.5	20
L5	6300	32	300
L6 (high stats)	3000	2000	40,000
L7 (high stats)	2800	20	400

• **PISA files (high stats)**

v01.00 - PISA files generated using oscNext high stats L7 v01.04 files.

v01.01 - Same as v01.00 but with additional variables relating to neutrino interactions (from `I3GENIEResultDict`) and the event header Julian date.

F.2 Processing times

Table 19 gives the processing times required by each stage of the selection. Note that only neutrino MC is currently shown (which dominates our total processing time, but at low levels detector data and background MC can be time consuming). The final level reconstruction (L6 for the high stats sample) dominates the processing time.

984 Appendix G Reconstruction comparison

985 In order to find the optimum reconstruction for the OscNext data selection, a detailed
986 apples-to-apples comparison of different low energy event reconstructions is needed. This
987 is addressed in the following sections.

988 The idea of a proper comparison entails that all reconstructions that are taken into
989 account run on the same files, within the same environment (hardware and software setup)
990 and their results are evaluated in the same way. Since the single reconstructions differ
991 strongly and were initially not available within the same hard- or software setup, all of
992 these points will be detailed below. In General, the task has been accomplished by includ-
993 ing the source code of the reconstructions considered in this comparison in the `oscNext-`
994 `metaproject` and with this setting up all reconstructions on cobalt. A number of common
995 and reconstruction-specific scripts allow any user to run any included reconstructions on
996 NPX and to evaluate and compare their output.

997 G.1 Evaluation criteria

998 Defining evaluation criteria that are (more or less) suitable for all different reconstructions
999 includes defining what parameters are to be compared as well as how these are quantified
1000 and visualized.

1001 Apart from the resolutions of the reconstructed interaction parameters, e.g. the com-
1002 putational cost of the single reconstructions is compared. These are measured as the run-
1003 time of the (possible) seed, of the reconstruction algorithm and of the entire reconstruction
1004 including all i/o and so on. Also, for likelihood-based reconstructions, the numbers of
1005 likelihood calls per reconstructed event are compared, since these (and the resulting table
1006 look-ups) are usually what drives the computation time.

1007 G.2 Common setup

1008 Within the setup, as color-coded in figure 35, there are common and reconstruction-specific
1009 scripts. The former include submitting jobs to the cluster, a wrapper around the actual
1010 job that is used for timing the total reconstruction run-time and evaluation and plotting
1011 scripts. The reconstruction-specific scripts are mostly the run-scripts calling the respective
1012 reconstruction within the job submitted to NPX.

1013 When the submit script is called with (as the minimum set of arguments) a gcd-file,
1014 input i3-file(s) and reconstruction name, it will submit one or the indicated number of
1015 jobs per input file (important for more time-consuming reconstructions). Each job calls
1016 the script wrapping timing and the run-script for the indicated reconstruction. Also, an
1017 evaluation job is submitted. This runs as the final job after all other jobs have terminated.
1018 Within this, the reconstructed values and other relevant data from the now reconstructed
1019 i3-files are written in a format common for all reconstructions. One output file now corre-

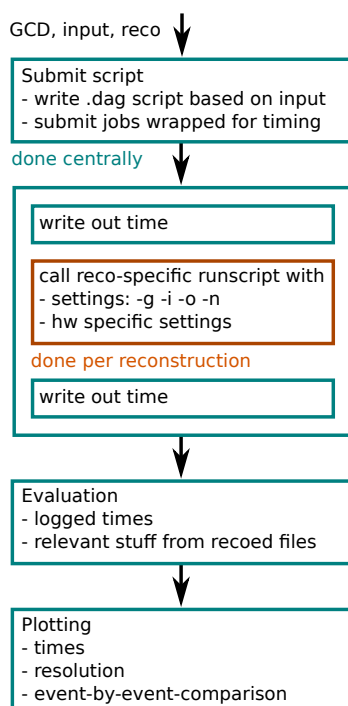


Figure 35: Structure of the reconstruction comparison framework setup on cobalt.

sponds to one call to the submit script and therefore to one set of settings with which one or a number of files have been reconstructed.

The plotting script takes one or more of such output files and can in this way compare different reconstructions or sets of settings for the same reconstruction.

All reconstructions that are mentioned within this comparison are or can be included in the OscNext-metaproject. They can be run using the corresponding repository³. A more technical set of instructions on how to use this setup can be found at the how to OscNext reco wiki⁴, together with instructions on how to include additional reconstructions.

G.3 Comparison

For comparison, different quantities and representations of those are used. The compared data set is a part (approx. three percent) of the OscNext-simulated muon neutrino events, processed to Level 5, v01.01. The code used for reconstructing are the oscNext-metaproject at revision 176759, the oscNextReco project at revision 177238 and retro-reconstructed files

³<http://code.icecube.wisc.edu/svn/sandbox/elohfink/oscNextReco/>

⁴https://wiki.icecube.wisc.edu/index.php/How_to_OscNext_reco

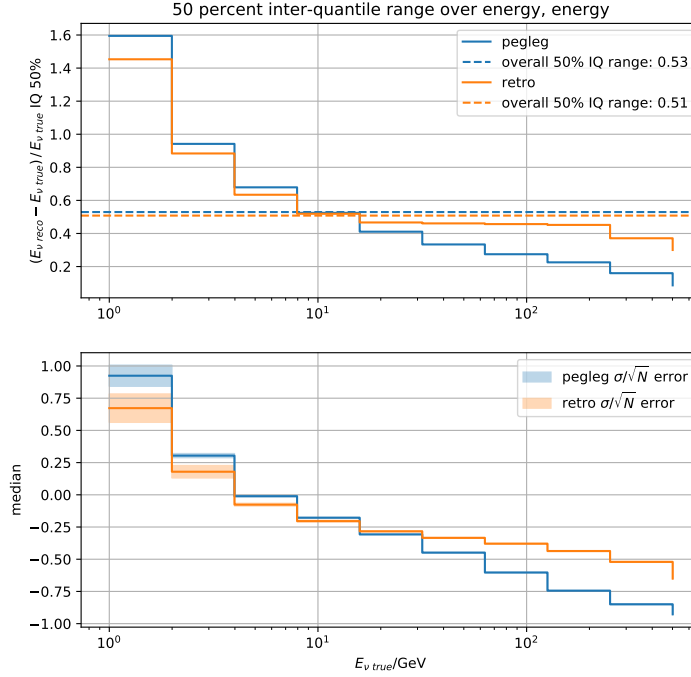


Figure 36: Reconstructed relative error of the total energy 50% interquantile ranges and median of the two main compared reconstructions over the logarithmic true energy. A comparison including SANTA and LEERA can be found in 43

1034 from

1035 Resolution

1036 The 50% interquantile range over the true energy is chosen as a measure for the resolution
 1037 as is shown in figures 36 and 37. Since this representation does not include systematic
 1038 errors, the median is always included as well. The interquantile ranges are the difference
 1039 between the 75% and the 25% quantiles, displayed as red solid lines in figure 39 and the
 1040 total mean interquantile range is shown as well in the overall error histogram in figure
 1041 40. For the x-axis, the true energy (and, secondarily, the number of hit DOMs, called
 1042 nChannel) are chosen since they yield the most consistent and relevant dependencies of the
 1043 resolutions in all parameters.

1044 In order to find possible populations of events that are reconstructed preferably with a
 1045 a non-default reconstruction, the error difference between two reconstructions is evaluated

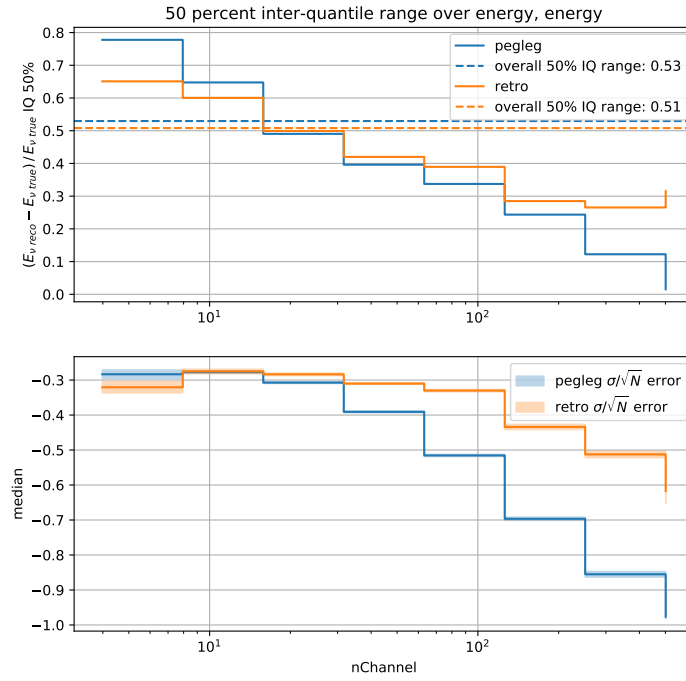


Figure 37: Reconstructed relative error of the total energy 50% interquantile ranges and median of the four compared reconstructions over the number of hit DOMs

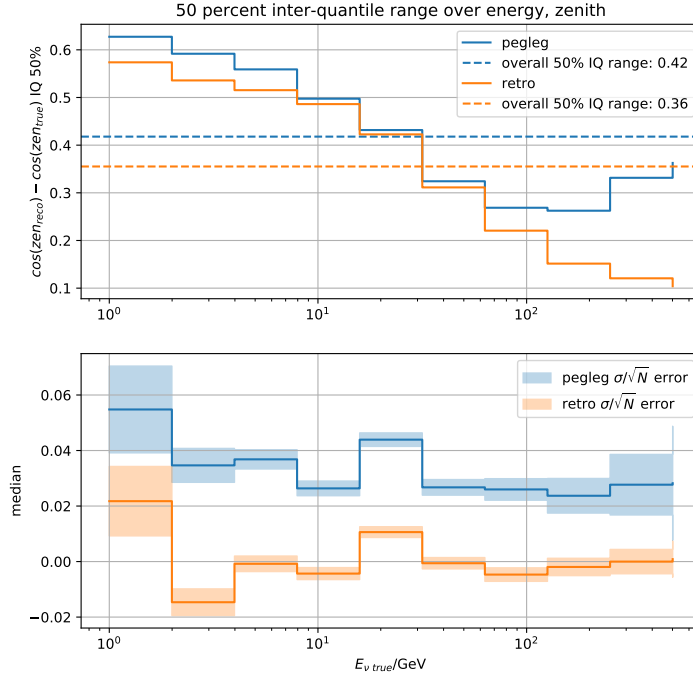


Figure 38: Reconstructed cosine zenith error 50% interquantile ranges and median of the two main compared reconstructions over the logarithmic true energy. A comparison including SANTA and LEERA can be found in 44

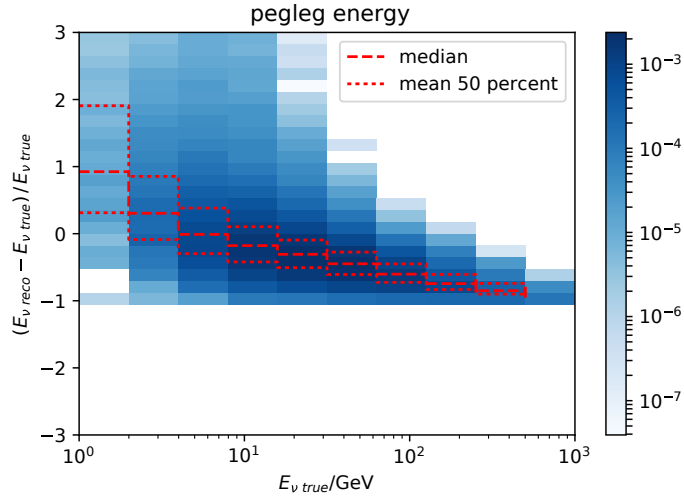


Figure 39: Relative energy error (true - reconstructed energy over true energy) over the logarithmic true energy. For all other parameters, the error instead of the relative error is shown. The color shows the number of events in the respective bin. The red lines show the median in the respective y-slice as well as the 25% and the 75% quantiles, which are used to compute the 50% interquartile range (figure 36).

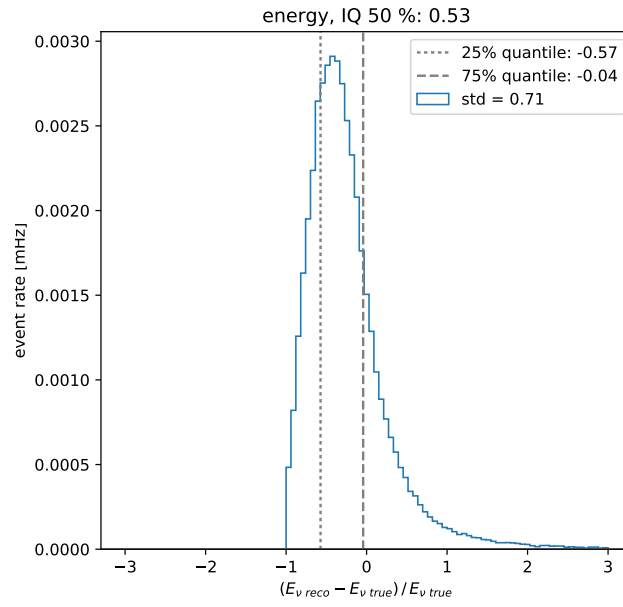


Figure 40: Histogram of the relative energy error in the entire data set. The overall interquantile range is used later, in figure 36. For all other parameters, the error instead of the relative error is used.

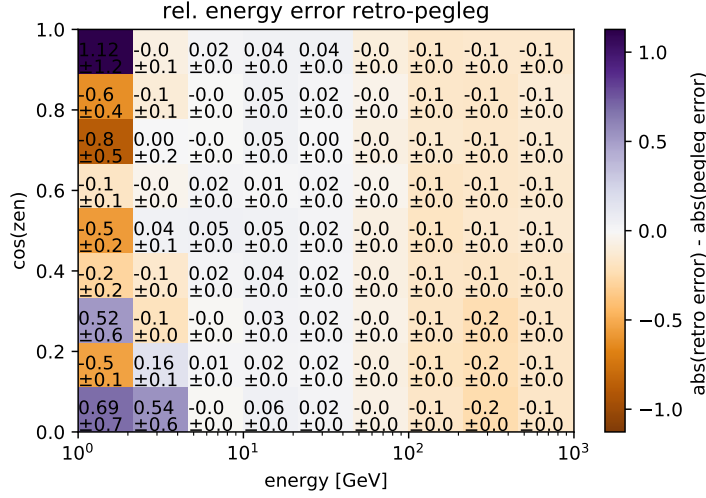


Figure 41: Difference between the absolute values of the reconstruction errors (usually true - reconstructed value; in this case the relative error, as usually used for the energy) of Retro and Pegleg, binned over the true $\cos(\text{zenith})$ and energy, in order to find populations where one reconstruction is preferred. This mainly reflects the known energy-dependence of the different performance of the two reconstructions.

as well. Since $\cos(\text{zenith})$ and energy are the most important parameters for low energy analyses, the differences of the absolute errors are shown over these two, as in figure 41. Within each bin, the mean value of the bin content as well as its standard error are noted, in order to clarify the meaningfulness of the bin content.

Run-time The run-time of the different reconstructions is compared in several ways: Where applicable, the number of likelihood evaluations is compared, as a proxy for the computational cost. This, however, does not apply to SANTA and LEERA, since these are not likelihood-based.

The run-time of just the reconstruction and seeding algorithm (separately) and the total run-time from starting to finishing of the job on the cluster are compared. The latter is included in order to get a realistic estimate of how long the reconstruction takes, where the first is rather the theoretical run-time. In figure 42, the reconstruction run-times for pegleg and retro are shown as well as the seed run-time for pegleg, which is not available for retro. These two are compared since they are the more time consuming reconstructions, being on an entirely different time scale than SANTA and LEERA.

In this context it needs to be noted that for running LEERA, a factor of three more jobs are submitted, accounting for the different steps that need to be taken (separately) in this reconstruction, the first basically being SANTA. Also, for Retro jobs, almost four

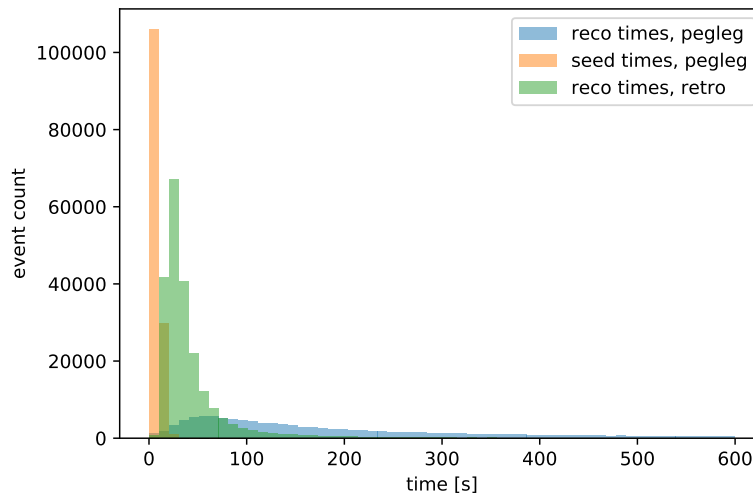


Figure 42: Run-time of the seeding (where available) and reconstruction algorithm. This does not include the overhead that is included in the total run-time.

1064 times the memory is needed for each job.

1065 **Efficiency** Other than the more complex reconstructions, which can safely be assumed
 1066 to reconstruct more than 99% of the events they are given, more simple reconstructions,
 1067 such as SANTA and LEERA, can only reconstruct events that fulfill certain criteria. This
 1068 is due to SANTA using only hits classified as un-scattered, direct hits and LEERA being
 1069 based on SANTA. As a result, these can only reconstruct approx. 34% of events.

1070 G.4 Conclusion

1071 Since the two modes of Retro are an improvement over the Pegleg performance in run-
 1072 time or resolution, respectively, and other complex reconstructions, such as DirectReco,
 1073 are currently not available for this kind of comparison, it seems straight-forward to have
 1074 Retro as the default reconstruction, in what ever mode fits the application best. Pegleg is
 1075 kept at the current level as a fall-back and reference reconstruction.

1076 G.5 Additional comparisons

1077

Where
did these
plots
come
from?
Justin?

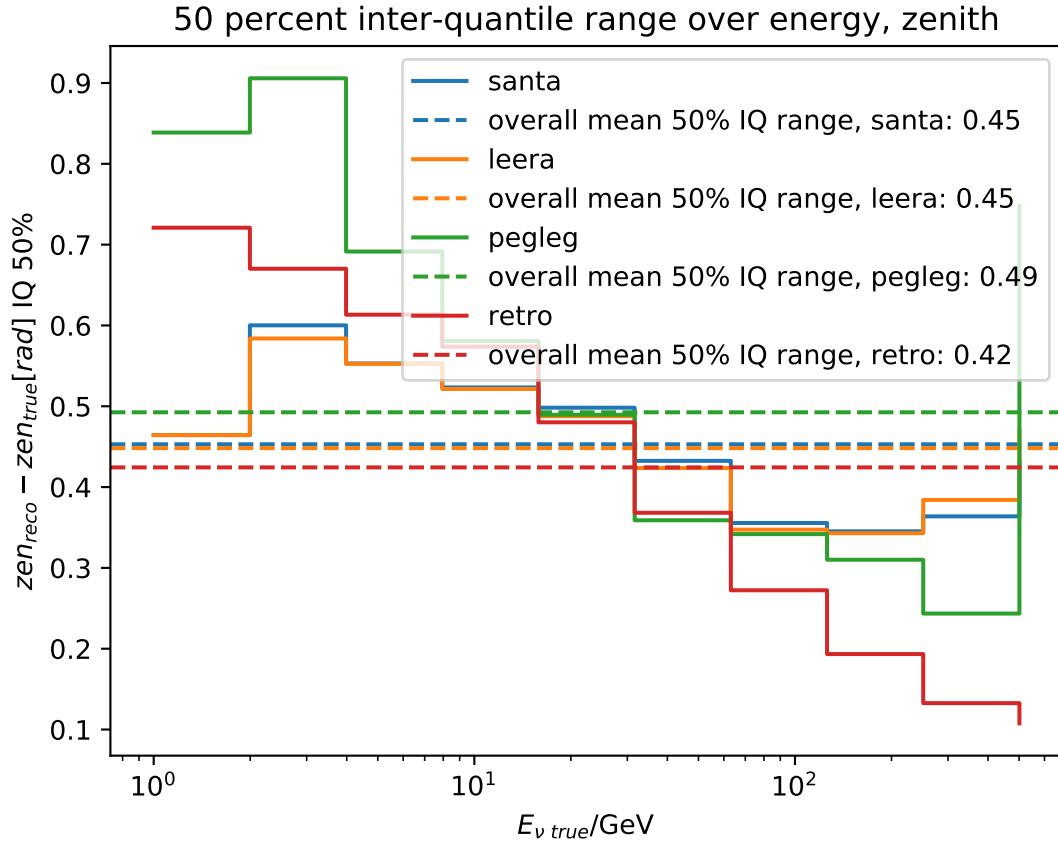


Figure 43: Reconstructed relative error of the total energy 50% interquantile ranges of all compared reconstructions over the logarithmic true energy.

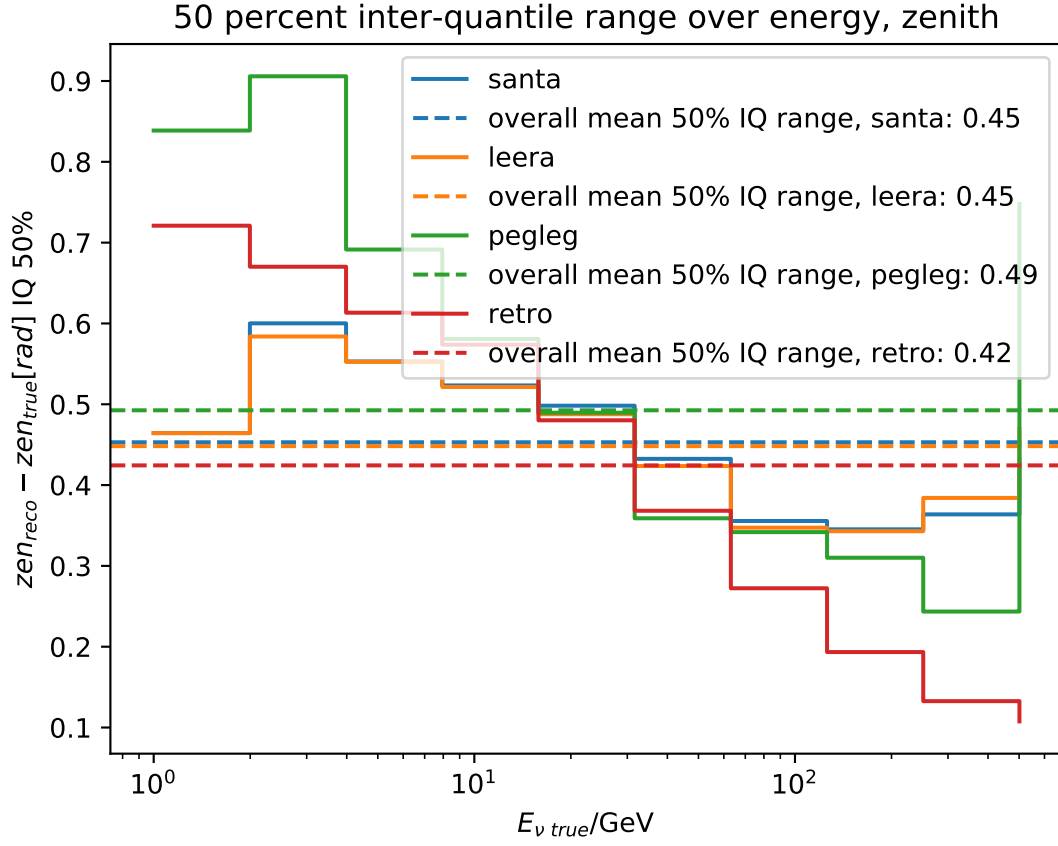


Figure 44: Reconstructed relative error of the total $\cos(\text{zenith})$ 50% interquantile ranges and median of all compared reconstructions over the logarithmic true energy.

Appendix H Known Issues

This is a list of known potential issues that have not yet been fully addressed:

- Some GENIE events are neither NC or CC (`InteractionType=0`). They appear to all be nue elastic scatters, but we currently don't treat these in PISA. Could lump them in with NC– but when we assume they are flavor independent this isn't quite correct for these events. Could lump them in with nueCC– but we need to check how cross section systematics are then handled in PISA.
- HitStatistics seems to fail in 0.1% of MuonGun files at L2 (+DC filter), not sure why. They are removed by L3 cut, but we're not sure what is going on.
- The L4 noise classifier is a bit overtrained
- Detector data rates at level 4 are lower for season IC86-17+ compared to the other years
- MuonGun only generates muons (not antimuons). It does this because in PROPOSAL there is not distinction between them in terms of energy loss, but we should check (a) whether there is a difference in energy loss expected for muons vs antimuons and (b) whether the calculated MuonGun flux weights correctly handles this
- Individual DOM single trigger rates are quieting over time. The decrease in SLC rate is visible in uncleaned pulses (`SplitInIcePulses`), at level 3 across the years
- Nominal and DOM efficiency simulation sets (GENIE and MuonGun) were not produced with the unified hole ice model. So we will need to make sure that we transform their parameters into their representation in the unified treatment when making hyperplanes
- We use a different hole ice model in the photon propagation (step 2) for GENIE (`as.flasher_p1.0.30_p2_-1`) vs MuonGun (`as.nominal`), meaning they have different effect DOM efficiencies
- Our method of making hole ice systematic sets by only re-running the detector sim with new hole ice parameterisations fails to take into account that we chose a particular hole ice parametrisation during photon propagation (which affects the effective DOM efficiency used in the sim)
- HQE doms have a different wavelength acceptance than regular doms, and this was corrected in CLSim with [this ticket](#) but that is not the version of CLSim that we are using. Unclear yet whether this is an issue for us.

1111 • The quasielastic and resonance axial mass systematic uncertainties are determined by
1112 varying the axial mass in GENIE at generation time and storing the cross section
1113 for each case. However, we test quite a narrow range here and the interpolation we
1114 use at analysis level based on this information can produce negative weights as the
1115 interpolation produces negative cross sections. We should test a wider range of axial
1116 mass values so we can properly map how the dependence of the cross section as it
1117 approaches zero.

References

- [1] Michael Larson. *Simulation and Identification of Non-Poissonian Noise Triggers in the IceCube Neutrino Detector*. Tech. rep. URL: <https://internal-apps.icecube.wisc.edu/reports/details.php?type=report%5C&id=icecube%5C%2F201312002> (cit. on p. 10).
- [2] Andrii Terliuk. *Adjustment of the vuvuzela noise parameters for simulations with the Pass2 SPE templates*. Tech. rep. URL: <https://www.overleaf.com/project/5d09feb0151adb57b602b36f> (cit. on p. 10).
- [3] M. G. Aartsen et al. “In-situ calibration of the single-photoelectron charge response of the IceCube photomultiplier tubes”. In: *Journal of Instrumentation* 15.6 (June 2020), P06032. DOI: [10.1088/1748-0221/15/06/P06032](https://doi.org/10.1088/1748-0221/15/06/P06032). arXiv: [2002.00997](https://arxiv.org/abs/2002.00997) (cit. on p. 10).
- (4) Costas Andreopoulos et al. “The GENIE Neutrino Monte Carlo Generator: Physics and User Manual”. In: (2015). arXiv: [1510.05494](https://arxiv.org/abs/1510.05494) (cit. on p. 14).